Judicial Case Scheduling in India

Thesis to be submitted in partial fulfillment of the requirements for the degree

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Dual Degree (B.Tech & M.Tech Integrated)

by

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CERTIFICATE

This is to certify that we have examined the thesis entitled **Judicial Case Scheduling in India**, submitted by **Mihir Mallick**(Roll Number: *21CS30031*) a postgraduate student of **Department of Computer Science & Engineering** in partial fulfillment for the award of degree of Dual Degree (B.Tech & M.Tech Integrated). We hereby accord our approval of it as a study carried out and presented in a manner required for its acceptance in partial fulfillment for the Post Graduate Degree for which it has been submitted. The thesis has fulfilled all the requirements as per the regulations of the Institute and has reached the standard needed for submission.

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In particular, the background motivation and several figures used in Chapter 1 draw from his earlier work. His thesis made the scheduling issues in the Indian judiciary concrete and accessible, enabling the present work to move further toward mathematical formulation, algorithmic scheduling experiments, and multi-objective analysis.

I am grateful for this foundational contribution, which served as the starting point for the direction and development of this thesis.

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ABSTRACT

The Indian judiciary faces persistent delays, repeated adjournments, and inefficient listing of matters, leading to long waiting times and uneven movement of cases through the system. While judicial delay is often discussed as a legal or institutional issue, it also has an operational scheduling dimension: every court working day, limited judicial capacity must be allocated among a much larger pool of pending matters. This thesis studies that problem through the lens of scheduling, focusing on arbitration-related matters from the Appellate Side of the Calcutta High Court.

Building on prior empirical work and curated case data, the thesis develops a feature-based abstraction in which each case is represented using scheduling-relevant attributes such as case age, inactivity gap, hearing-history count, order count, and case type. It formulates case selection under limited listing capacity and evaluates deterministic, interpretable policies including oldest-first, gap-first, weighted age-gap, balanced priority, shortest-job-first proxy, and round-robin case-type scheduling.

The experiments extend to coefficient sensitivity, bi-objective scheduling, and multi-objective scheduling across delay priority, case-type fairness, and complexity control. The results show that delay-oriented policies select old and inactive matters effectively, but may underrepresent smaller case categories; fairness-aware policies improve representation but can reduce delay-priority quality; and complexity-aware policies select simpler matters but may leave older cases behind.

These findings indicate that judicial case scheduling could be treated as a multi-objective decision problem rather than a single-score ranking task. The thesis presents a transparent decision-support framework for exposing scheduling trade-offs and motivates future work on stochastic hearing simulation, judge and advocate constraints, prediction-assisted scheduling, genetic algorithms, and reinforcement learning.

Keywords: judicial scheduling, case listing, Indian judiciary, arbitration cases, court delay, multi-objective optimisation

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Chapter 1

Introduction

1.1 Background and Motivation

The administration of justice is often discussed in legal, constitutional, and institutional terms. However, at the operational level, a court is also a large service system. Every working day, a finite number of judges and benches must allocate limited hearing time among a much larger pool of pending matters. The daily cause list is therefore not merely a clerical document; it is the visible output of a scheduling decision. It determines which matters receive judicial attention, which matters wait, and which parts of the pending stock continue to age.

This perspective is important because delay in the judiciary is not produced only by the number of pending cases. Delay is also shaped by how listing opportunities are distributed. If the cause list is overloaded, many matters may be listed but not reached. If a listed matter is not procedurally ready, the listing may not produce useful progress. If the same lawyer or party is required in several matters on the same day, at least some of those listings may become ineffective. If the system repeatedly prioritises a narrow class of matters, other case types may experience starvation even when the aggregate number of listed cases appears high. These are scheduling failures in addition to being administrative or legal difficulties.

The problem is further complicated by the fact that court cases are not homogeneous jobs. They differ in filing age, case type, statutory section, urgency, number of past hearings, last-hearing gap, stage of proceeding, bench requirement, and probability of disposal. A matter may be listed on a particular day but the outcome of that listing may be hearing, adjournment, absence, dismissal, withdrawal, or simply "not today" because the court did not have enough effective time to reach it. Hence, the core operational issue is not only how many matters are listed, but which matters are listed and whether that listing is likely to reduce delay.

The present thesis studies this problem through the lens of scheduling and optimisation. It focuses on arbitration-related cases, especially from the Calcutta High Court, and develops a mathematical and experimental framework for evaluating alternative listing policies. The central question is: given a finite listing capacity and a set of pending cases, how should the system select cases if it wishes to balance age reduction, inactivity reduction, throughput, and fairness across case categories?

1.2 Prior Empirical Basis and Positioning of This Thesis

This thesis builds on the earlier work of Avishek Mukhopadhyay on automated scheduling of cases in India. Unless explicitly stated otherwise, the empirical figures and court-level statistics in this introductory chapter are reproduced or adapted from Avishek Mukhopadhyay's 2021 MTP work and associated presentation materials. They are used as historical and diagnostic background for the present thesis, not as newly collected current-year statistics. The present thesis builds on this empirical foundation by formalising the scheduling problem mathematically and experimentally evaluating scheduling policies on the curated arbitration case dataset. That work performed the essential empirical groundwork: it collected cause-list and case-detail data, automated portions of the data-collection process, visualised delay patterns across courts, and made the scheduling problem visible through court-level and arbitration-specific plots (Mukhopadhyay 2021*a*,*b*). In particular, the earlier work examined the Supreme Court and multiple High Courts, including Calcutta, Chhattisgarh, Orissa, Rajasthan, Punjab, and Sikkim, along with selected district or taluka courts. It also moved from general court-hearing analysis to a deeper study of arbitration cases.

The contribution of Mukhopadhyay's work may be understood as empirical and diagnostic. It established that the problem is observable in real court data, identified repeated symptoms of delay, and suggested practical remedies such as early screening, avoiding listing when stakeholders are absent, tracking time-limit violations, limiting high-priority matters per day, distributing cases among available judges, preferring division benches where appropriate, and capping excessive hearings per case. However, those suggestions were not developed into a formal optimisation model or experimentally evaluated as scheduling policies.

The present thesis starts from those empirical observations but moves the work in a different direction. Its contribution is to formalise the problem mathematically, define scheduling-relevant state variables, construct interpretable policies, run controlled experiments, and compare the behaviour of those policies using quantitative metrics. In other words, the earlier work answers the question: *what patterns of delay are visible in the data?* This thesis asks the next question: *if these patterns are treated as a scheduling problem, how do different listing rules behave and what trade-offs do they reveal?*

This distinction is important for the structure of the report. The figures in this chapter are adapted from Mukhopadhyay's data collection and visualisation work and are used as empirical motivation. The modelling, mathematical abstraction, objective design, algorithmic policy implementation, and experimental interpretation developed in the later chapters constitute the original contribution of this thesis.

1.3 Court Scheduling as an Operational Problem

Mukhopadhyay's initial problem analysis identifies several practical causes of scheduling difficulty. The first is manual or semi-manual case scheduling: the daily list of cases for hearing is often decided based on directions, administrative practice, and available information rather than by an explicit optimisation model. The second is stakeholder uncertainty: judges, lawyers, or parties may be absent. The third is stakeholder conflict: the same participant may be involved in multiple cases and may be unable to appear at multiple places on the same day. The fourth is priority imbalance: long-running matters can be postponed when other matters receive higher priority.

These observations suggest that court scheduling should not be treated as a simple first-in-first-out queue. A court listing system is closer to a constrained, multi-stage, stochastic scheduling system. The resource is not just a generic server; it is a bench with legal compatibility, roster constraints, and finite daily capacity. The job is not just a unit task; it is a case that may require repeated hearings before disposal. The service outcome is not deterministic; even after listing, the matter may fail to progress.

The earlier data-collection framework also clarifies what information is relevant for such a model. Cause lists contain court and listing information such as court number, case number, advocate details, party details, district name, date, and time. Case-detail pages contain case number, case age, number of hearings already completed, case type, and linked-case information. These fields correspond closely to scheduling state variables: age, hearing count, case type, participant involvement, and linkage constraints.

Thus, the thesis begins from a practical observation: the judiciary already produces the raw ingredients required for scheduling analysis. What is missing is a formal layer that converts those ingredients into state variables, objectives, constraints, and policies.

1.4 Cross-Court Evidence of Delay

The broader empirical study examined delayed and on-time hearing behaviour across multiple courts. This is important because it shows that the scheduling issue is not isolated to a single dataset. Across the Supreme Court and several High Courts, the earlier analysis recorded large numbers of delayed or not-taken-up hearings, often with "Not Today" as a dominant reason. Figure 1.1 presents the broad delayed-versus-on-time hearing pattern across courts.

The court-wise numbers reported in the earlier work show the scale of the issue. For the Supreme Court, the August 2022 analysis reported 11,517 cause-list entries, 3,845 delayed or not-taken-up hearings, and 7,781 on-time hearings. For the Calcutta High Court, it reported 35,902 cause-list entries, 13,462 on-time hearings, and 22,440 delayed or not-taken-up hearings. For Chhattisgarh High Court, it reported 19,046 entries, with 10,597 on-time and 8,449 delayed or not-taken-up hearings. For Orissa High Court, it reported 51,939 entries, with 23,182 on-time and 28,757 delayed or not-taken-up hearings. For Rajasthan High Court, it reported 2,613 entries, with 1,365 on-time and 1,248 delayed or not-taken-up hearings.

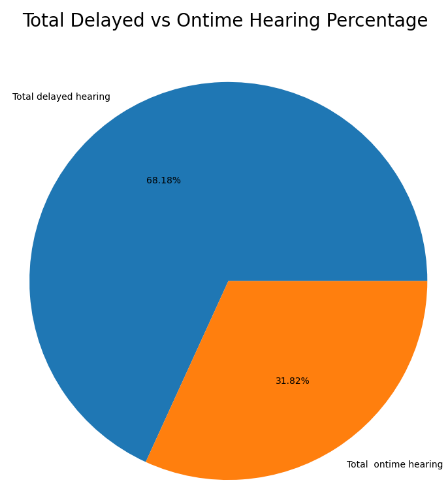

Figure 1.1: Delayed and on-time hearing percentage across courts. Source: adapted from Mukhopadhyay's 2021 automated scheduling analysis.

Table 1.1: Court-wise hearing statistics reported in the prior empirical analysis.

Court	**Cause-list entries**	**On-time hearings**	**Delayed/not taken up**
Supreme Court	11517	7781	3845
Calcutta High Court	35902	13462	22440
Chhattisgarh High Court	19046	10597	8449
Orissa High Court	51939	23182	28757
Rajasthan High Court	2613	1365	1248

The Calcutta numbers are especially striking because delayed or not-taken-up hearings form the majority of reported cause-list outcomes. This does not by itself prove that Calcutta is uniquely inefficient, because the courts differ in case mix, reporting conventions, and time period. However, it does show that Calcutta provides a meaningful setting for studying delay as an operational scheduling phenomenon.

The main implication of these numbers is that the cause list cannot be assumed to be equivalent to actual hearing progress. A matter may appear on the list but fail to move forward. Therefore, any realistic abstraction must separate the *listing decision* from the *execution outcome*. This distinction becomes central in Chapter 2, where the system is modelled as a scheduling process with uncertain hearing outcomes.

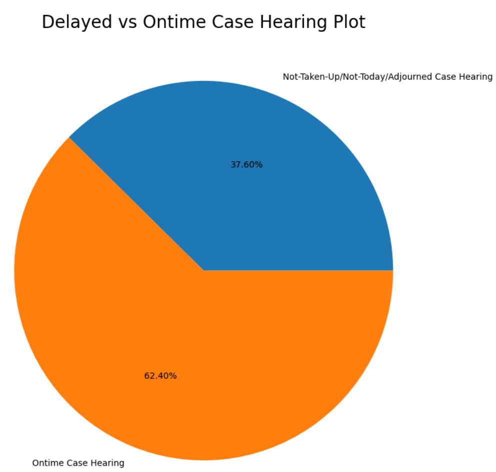

Figure 1.2: Delayed and on-time hearing split for Calcutta High Court cause-list entries. Source: adapted from Mukhopadhyay's 2021 analysis.

1.5 Arbitration Cases

After the general court-level analysis, Mukhopadhyay's work identifies arbitration matters as a useful category for deeper study. The earlier presentation notes that arbitration cases were among the top delayed case types and that this observation was cross-checked through interaction with lawyers from different law firms. This makes arbitration a natural domain for a scheduling thesis because it is both operationally significant and legally structured.

Arbitration is intended to provide an efficient mechanism for dispute resolution, where parties submit their dispute to one or more arbitrators who make a binding decision. However, courts remain involved in several arbitration-related stages, including interim relief, appointment of arbitrators, removal or substitution of arbitrators, enforcement, and challenges to awards. Court delay in such matters can therefore weaken the efficiency promise of arbitration itself.

The arbitration dataset used in the earlier work spans 11 years, from 2012 to 2022, for multiple courts. For Calcutta High Court, the arbitration data table reports 10,603 case-hearing records and 7,308 distinct cases. Comparable data were also collected for Orissa High Court and Jaipur/Jodhpur-related courts. The important point for this thesis is not merely the volume of data but the structure of the problem: arbitration cases repeatedly generate hearing demands, vary widely in time to disposal, and differ by section and bench requirement.

This thesis focuses especially on Calcutta arbitration matters for three reasons. First, the Calcutta High Court data show a large delayed-hearing component in the broader analysis. Second, the arbitration data provide enough historical depth to study case age, hearing count, and delay patterns. Third, the case types available in the current project data, including categories such as FMAT, AOCOM, and ADCOM, allow scheduling policies to be evaluated not only by age or gap but also by fairness across case categories. Thus, Calcutta arbitration matters are not chosen arbitrarily: they arise from the prior empirical observation that arbitration-related matters appeared among the delayed categories, and from the availability of long-horizon, case-level arbitration data that allows age, hearing-count, bench-type, and case-type behavior to be studied.

1.6 Calcutta Arbitration: Empirical Situation

The arbitration-specific plots show why a simple scheduling rule is unlikely to be sufficient. Figure 1.3 compares delayed and on-time arbitration cases in the Calcutta arbitration data. The dominance of delayed cases indicates that delay is not a small tail event. It is a structural feature of the case population under study.

This figure motivates the first operational question of the thesis: if a court cannot list all pending matters, which delayed cases should receive priority? An oldest-first policy would focus on case age. A gap-first policy would focus on time since last hearing. A hearing-count-aware policy would account for repeated listing history. A case-type-aware policy would protect categories that may otherwise be underrepresented. The rest of the thesis develops and compares such alternatives.

1.6.1 Repeated hearings and the long tail

A central feature of arbitration matters is that they do not behave like one-time jobs. Figure 1.4 shows the cumulative distribution of hearing count. The curve rises quickly for matters with few hearings, but it also contains a long tail of cases that require many hearings. Mukhopadhyay's analysis notes that roughly 50–60% of arbitration cases are disposed on the first date. However, if first-day disposed matters

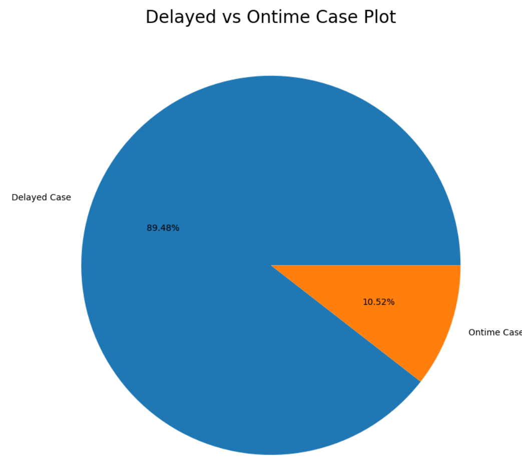

Figure 1.3: Delayed versus on-time arbitration cases in the Calcutta arbitration data. Source: adapted from Mukhopadhyay's 2021 arbitration case analysis.

are ignored, many remaining cases require around 10 to 15 hearings and 150 to 200 days to complete. Some matters have more than 100 hearings and more than 1200 days of age.

This long-tail behaviour has a direct scheduling implication. If a policy focuses only on easy or low-hearing-count cases, it may improve short-term completion indicators while leaving long-running matters unresolved. If a policy focuses only on high-hearing-count or old matters, it may reduce extreme backlog but consume capacity on difficult cases. Therefore, hearing count is not merely a descriptive field; it is a scheduling signal.

1.6.2 Case age and delay accumulation

Case age captures the litigant-facing burden of delay. Figure 1.5 shows the cumulative distribution of disposed arbitration case age. The shape of this curve is important because it reveals that disposal and delay are different quantities. A case may eventually be disposed, but the time taken to reach that disposal can still be high. Therefore, a scheduling policy must be evaluated not only by the number of cases selected or disposed but also by the age profile of those cases.

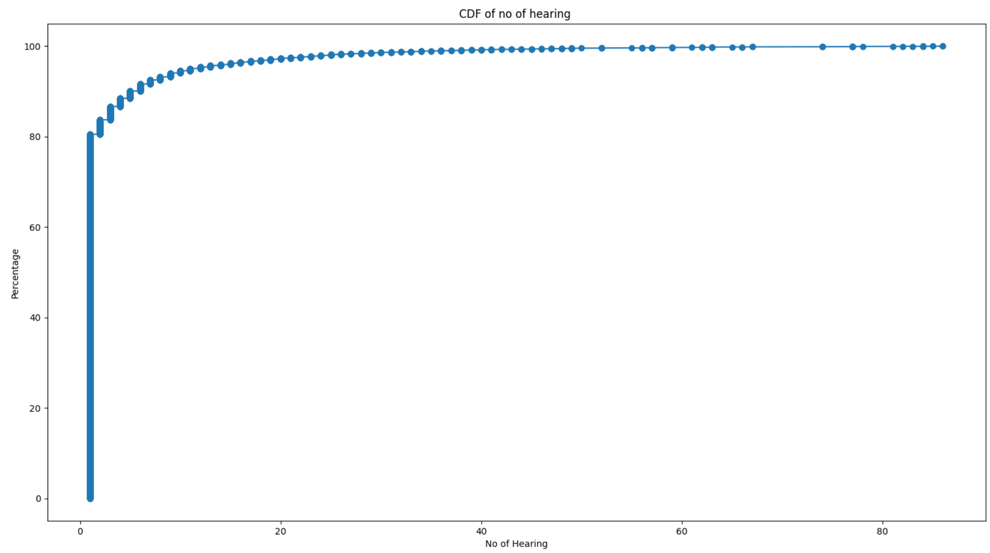

Figure 1.4: Cumulative distribution of the number of hearings per arbitration case. Source: adapted from Mukhopadhyay's 2021 arbitration case analysis.

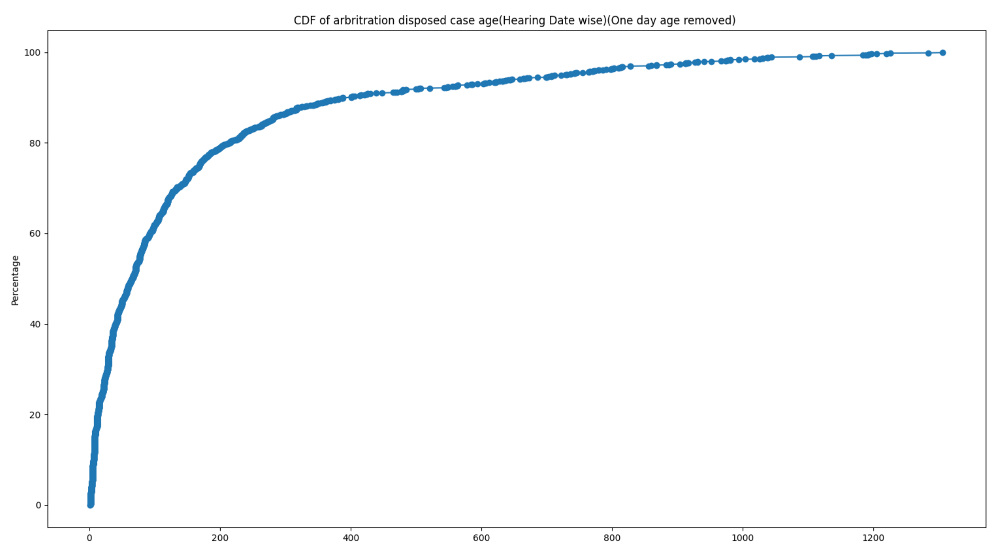

Figure 1.5: Cumulative distribution of disposed arbitration case age using hearing-date-based age. Source: adapted from Mukhopadhyay's 2021 arbitration case analysis.

The current thesis uses case age as one of the core state variables. It also uses age directly in the single-objective experiments, where the FCFS or oldest-first policy selects the oldest available cases. However, as later results show, an age-only policy is not enough: it improves one dimension of delay but may not produce balanced treatment across case categories.

1.6.3 Section-sensitive delay

Delay is not uniformly distributed across all arbitration-related legal sections. Figure 1.6 shows delayed cases by Arbitration Act section. The earlier analysis specifically highlights Sections 9 and 11 as delay-prone. Section 9 is associated with interim measures, while Section 11 concerns appointment of arbitrators. Both may involve repeated court interaction and therefore may produce distinct scheduling demands.

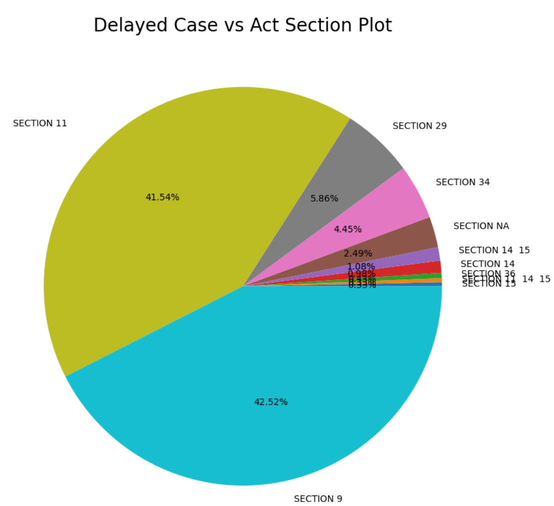

Figure 1.6: Delayed arbitration cases by Act section. Source: adapted from Mukhopadhyay's 2021 arbitration case analysis.

The scheduling implication is that a policy should not be judged only by aggregate averages. A policy may improve the average selected age while still ignoring a case type or legal section. This is why the current thesis later includes round-robin and balanced-priority policies that explicitly address category representation.

1.6.4 Bench dependence and single point of failure

Bench structure is another important feature. The earlier arbitration analysis distinguishes between single-bench and division-bench matters and notes that single-bench cases are mostly delayed. It further observes that approximately 99% of cases in the studied arbitration data have single-bench handling, creating a single point of failure. Figure 1.7 illustrates this distribution.

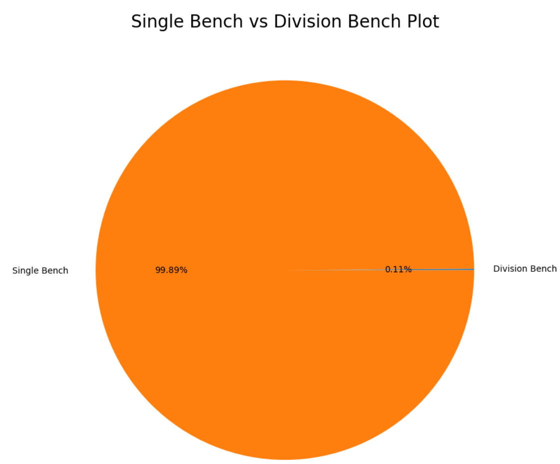

Figure 1.7: Single-bench and division-bench distribution in arbitration matters. Source: adapted from Mukhopadhyay's 2021 arbitration case analysis.

This motivates bench compatibility and bench availability constraints in the formal model. If a case can only be heard by a particular bench type, scheduling cannot be modelled as selection from a single unconstrained pool. Even if the first implementation abstracts away detailed bench assignment, the full formulation must leave space for this constraint.

1.6.5 Judge and advocate concentration

The earlier analysis also shows concentration in the distribution of matters across judges and advocates. It notes that very few judges handled a majority of arbitration cases and that only 45 judges handled arbitration cases at the Calcutta High Court over the 11-year period. It also notes that few advocates handled a majority of cases, suggesting concentration on the participant side as well.

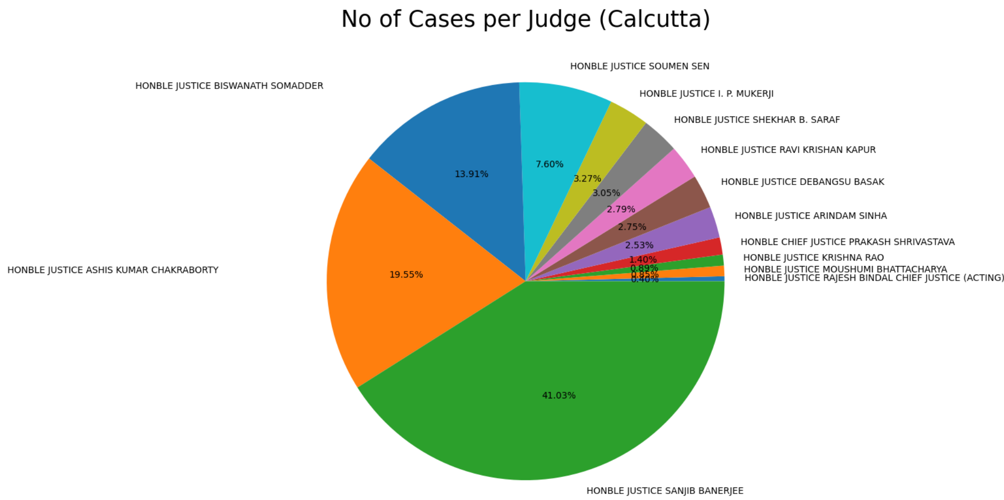

Figure 1.8: Distribution of arbitration case hearings across judges in the Calcutta data. Source: adapted from Mukhopadhyay's 2021 arbitration case analysis.

These observations are important because a realistic schedule must eventually handle resource conflicts. A case consumes not only a listing slot but also the time of a bench and the availability of lawyers or parties. The current thesis begins with a simpler selection-level abstraction, but the full modelling direction includes these resources explicitly.

1.7 From Practical Remedies to Mathematical Policies

Mukhopadhyay's work concludes with several practical remedies. These include initial screening before case registration, not scheduling cases when any required stakeholder is absent, tracking whether a case has violated time limits, setting a maximum limit on arbitrator changes, tracking arbitrator performance, limiting high-priority cases per day, dividing cases among available judges, allocating division benches rather than relying only on single benches, setting a maximum number of cases taken by

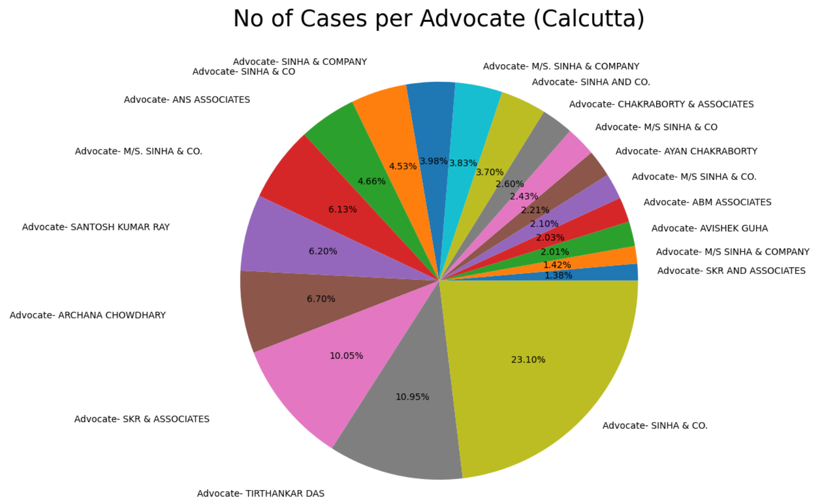

Figure 1.9: Distribution of arbitration case hearings across advocates. Source: adapted from Mukhopadhyay's 2021 arbitration case analysis.

an advocate, setting a maximum number of hearings per case, and recruiting more judges.

These recommendations are useful because they identify plausible intervention points. However, they remain qualitative unless converted into measurable objectives and constraints. This thesis performs that conversion. For example, "limit high-priority cases per day" becomes a capacity or quota constraint. "Set maximum number of hearings per case" becomes a hearing-count penalty or intervention threshold. "Divide cases among available judges" becomes a load-balancing constraint. "Do not schedule if a stakeholder is absent" becomes a feasibility constraint. "Track time limits" becomes an age or deadline-based priority term.

The mathematical transformation is the main bridge from Chapter 1 to Chapter 2. In Chapter 2, a case is represented as a state vector containing age, case type, last-hearing gap, hearing count, and other relevant features. A scheduling policy becomes a rule or optimisation procedure that selects a subset of cases subject to capacity. The policies are then compared experimentally in Chapter 3.

1.8 Problem Statement

The central problem studied in this thesis can be stated as follows:

> Given a pool of arbitration-related matters waiting for judicial attention, and given that only a limited number of matters can be listed or effectively prioritised on a court day, how should the system choose which cases to take up so that long-pending and long-inactive matters are not ignored, while smaller case categories are not starved and the selected schedule remains practically manageable?

This statement captures the operational difficulty behind court listing. A court does not merely face a large number of cases; it faces a repeated daily choice under limited capacity. If every pending matter could be heard immediately, scheduling would not be a meaningful problem. The problem arises because only a subset of matters can be listed, and the choice of that subset determines which cases move forward and which cases remain in the backlog.

The empirical observations discussed earlier show that delay has multiple dimensions. A case may be old in terms of total age, or it may have remained inactive for a long period since its last hearing or order. A matter may also have a long hearing history, indicating that it has repeatedly returned to the system without final resolution. At the same time, the data contains different arbitration-related case categories such as FMAT, AOCOM, and ADCOM. A policy that selects only the oldest or highest-delay cases may appear efficient but may still underrepresent smaller categories. Therefore, the problem is not simply to find the "most delayed" cases, but to understand the trade-off between delay reduction, inactivity reduction, case-type representation, and schedule complexity.

This thesis approaches the problem by treating listing as a case-selection task. Each case is represented by scheduling-relevant features such as case age, time since last activity, hearing-history count, order count, and case type. A scheduling policy observes these features and selects a limited set of cases for prioritisation. Different policies encode different administrative intuitions: oldest-first selection gives priority to the longest pending matters, gap-first selection targets inactivity, weighted priority combines multiple delay indicators, shortest-job-first-style selection favours simpler matters, and round-robin selection improves representation across case types.

The purpose of this formulation is not to replace judicial discretion or to claim that legal priority can be reduced to a numerical score. Rather, the aim is to build a controlled experimental framework in which different listing philosophies can be compared. Once the problem is expressed in terms of case features, capacity, and measurable outcomes, it becomes possible to ask precise questions: Which policy selects the oldest matters? Which one reduces long inactivity? Which one leaves fewer old cases behind? Which one gives better representation to smaller case categories? Which one sacrifices delay priority in order to improve fairness or reduce complexity?

Thus, the problem statement naturally leads to the mathematical framework developed in the next chapter. Chapter 2 formalises the case features, decision variables, constraints, and objective functions used to study this scheduling problem. Chapter 3 then uses that framework to compare single-objective, bi-objective, and multi-objective scheduling policies on the arbitration case dataset.

1.9 Research Objectives

The objectives of this thesis are designed to move systematically from empirical observation to formal scheduling and then to experimental evaluation. They are as follows:

1. **To formulate court listing as a scheduling problem.** The first objective is to reinterpret arbitration case listing as a capacity-constrained scheduling problem in which pending cases compete for limited listing slots.

2. **To construct scheduling-relevant case features.** The thesis uses case age, inactivity gap, hearing-history count, order count, and case type as the main operational features. These variables are measurable from the available case records and directly correspond to delay, inactivity, repeated listing, activity complexity, and case-category representation.

3. **To design interpretable single-objective policies.** The work compares a family of transparent scheduling policies: oldest-first or FCFS selection, gap-first selection, weighted age-gap priority, balanced priority with mild complexity penalty, shortest-job-first proxy, and round-robin case-type selection. Each policy represents a different administrative priority.

4. **To test coefficient sensitivity.** Since weighted scores depend on the chosen coefficients, the thesis studies age-dominant, gap-dominant, hearing-history-dominant, balanced, and complexity-penalised profiles. This checks whether the selected schedules are robust to moderate changes in the scoring function.

5. **To evaluate bi-objective trade-offs.** The thesis studies delay priority against case-type fairness, and delay priority against complexity. This identifies whether improving representation or reducing complexity substantially changes which matters are selected.

6. **To evaluate an integrated multi-objective policy.** The final experimental layer combines delay priority, case-type fairness, and complexity control using an interpretable greedy scalarisation. This gives a practical approximation to the multi-objective nature of judicial scheduling without immediately moving to black-box optimisation.

7. **To identify limitations and future directions.** The thesis uses the experimental findings to motivate richer future methods such as genetic algorithms, NSGA-II, simulated annealing, and reinforcement learning, but only after establishing clear rule-based and optimisation-based baselines.

The central research contribution is therefore not merely proposing that old cases should be prioritised. Rather, it is the systematic comparison of multiple scheduling philosophies under the same data representation and capacity assumptions.

1.10 Scope and Assumptions

The scope of this thesis is intentionally limited to a tractable first formalisation of judiciary scheduling. The empirical focus is on arbitration-related matters from the Calcutta High Court, represented through the case types FMAT, AOCOM, and ADCOM. The experiments operate on a curated case pool and compare how different policies select cases from that pool.

The following assumptions define the boundary of the present work:

1. **Selection-level scheduling.** The thesis studies which cases should be selected for listing, not the exact intra-day ordering of matters before a bench.

2. **Fixed listing capacity.** Each experiment assumes a fixed capacity, such as 10, 20, or 30 selected matters. This approximates limited daily judicial attention and allows capacity sensitivity to be studied explicitly.

3. **Case-level independence.** Cases are treated as independent scheduling units unless explicit linkage information is available. Complex dependencies between connected matters are left for future work.

4. **Feature-based priority.** The policies use measurable features such as age, inactivity gap, hearing-history count, order count, and case type. The model does not attempt to infer legal merits, urgency from pleadings, or judge-specific directions.

5. **Simplified backlog evolution.** In the multi-day simulation, selected cases are removed from the immediate listing backlog as a first-phase approximation. A richer simulator would model whether a listed case is actually heard, adjourned, disposed, or returned to the queue.

6. **No explicit judge or advocate availability model.** The current experiments do not fully model judge rosters, advocate conflicts, stakeholder absence, or bench-specific compatibility. These constraints are recognised in the abstraction but are not optimised in the first implementation.

7. **Fairness as case-type representation.** Fairness is measured through representation of FMAT, AOCOM, and ADCOM in the selected set relative to the available pool. This is a scheduling fairness proxy, not a complete legal fairness criterion.

8. **Comparative, not prescriptive, results.** The purpose of the experiments is to expose trade-offs between scheduling rules. The thesis does not claim that any one policy can be directly deployed without institutional validation, legal review, and richer operational constraints.

These assumptions keep the model transparent and experimentally manageable. They also create a clear path for extension: later versions can add stochastic hearing outcomes, judge and advocate availability, bench compatibility, observed hearing durations, disposal probability, and richer optimisation algorithms.

1.11 Contribution of the Thesis

The contribution of this thesis is best understood relative to the prior empirical work. Mukhopadhyay's work collected and visualised data, identified recurring delay patterns, and suggested practical directions. This thesis takes those observations as input and develops the next layer: mathematical operationalisation and experimental policy evaluation.

The specific contributions are:

1. A formal scheduling interpretation of court-hearing management, where pending cases are modelled as schedulable entities and listing capacity is treated as a constrained resource.

2. A case-selection abstraction for Calcutta arbitration matters using age, inactivity gap, hearing count, and case type as scheduling features.

3. A set of interpretable scheduling policies corresponding to different administrative objectives.

4. A controlled experimental framework for comparing these policies under common data and metrics.

5. An interpretation of the resulting trade-offs, showing why no single-objective policy fully captures the requirements of judicial scheduling.

This framing ensures that the thesis is not merely a reproduction of earlier plots. The earlier plots motivate the problem; the present work formalises, implements, solves, and analyses scheduling policies derived from that problem.

1.12 Thesis Organisation

The remainder of the thesis is organised as follows. Chapter 2 develops the abstraction and mathematical formulation of the judiciary scheduling problem. It defines the entities, state variables, objectives, assumptions, and policy families used in the experiments. Chapter 3 presents the experimental runs, visualisations, and results. It compares how single-objective scheduling policies behave and explains their trade-offs. Chapter 4 discusses future work, including multi-objective optimisation, genetic algorithms, richer stochastic simulation, and reinforcement learning. Chapter 5 concludes the thesis by summarising both the mathematical insights and the social implications of scheduling-aware judicial administration.

The next chapter therefore moves from empirical motivation to formal modelling. The observations in this chapter show that delay is real, repeated, and structured. Chapter 2 turns those observations into variables, constraints, objectives, and algorithms.

Chapter 2

Abstraction and Formulation

2.1 From Court Practice to Scheduling Abstraction

The previous chapter established the empirical motivation for studying court listings as a scheduling problem. The central observation was that judicial delay is not only a matter of total case volume. It is also affected by how listing opportunities are distributed among pending matters. A cause list is therefore not simply an administrative record; it is the outcome of repeated selection decisions made under limited judicial capacity.

This chapter develops the abstraction used in the rest of the thesis. The aim is not to model the entire legal and administrative machinery of the High Court. Such a model would require detailed information about judge rosters, advocate availability, urgency directions, court-specific practices, procedural readiness, linked matters, adjournment reasons, and hearing durations. Instead, the objective here is to construct a minimal but expressive formulation that captures the part of the system relevant to the experiments performed in this thesis: selecting a limited number of arbitration matters from a larger case pool.

The abstraction begins with a simple idea. On a given scheduling day, the court has a set of cases that may be considered for listing. Each case has observable features such as age, time since last activity, previous hearing history, order history, and case type. Because the court cannot list or effectively prioritise every matter at once, a scheduling policy selects a subset of cases. The quality of this subset is evaluated using measurable criteria: whether the selected matters are old, whether they have been inactive for long, whether repeated-hearing matters are being addressed, whether smaller case categories are represented, and whether the selected set is excessively complex.

This abstraction is intentionally placed between two extremes. At one extreme, it would be too simplistic to say that the court should always hear the oldest case first. Such a rule ignores inactivity gap, case type, repeated hearings, and fairness across categories. At the other extreme, it would be unrealistic for this thesis to claim a complete end-to-end simulator of court functioning. The present formulation therefore keeps the model focused: it studies case selection under capacity, while making explicit which richer features are left for future work.

2.2 Modelling Choice: Selection-Level Scheduling

Court scheduling may be modelled at different levels of detail. A minute-level model would decide the exact intra-day order of hearings before a bench. A bench-assignment model would decide which case goes before which bench or judge. A stochastic execution model would decide whether a listed case is actually heard, adjourned, not reached, or disposed. These are important modelling directions, but they require more granular data and stronger assumptions.

The present thesis uses a selection-level model. The scheduler is not asked to decide the exact clock time of a hearing. It is also not asked to simulate the detailed courtroom outcome of every listed matter. Instead, the scheduler answers the following question:

> Given a set of available arbitration matters and a fixed listing capacity, which matters should be selected for priority listing under a given scheduling rule?

This level of abstraction is appropriate for the current work for three reasons. First, the available dataset supports reliable construction of case-level features such as age, inactivity gap, hearing history, order count, and case type. Second, the selection problem is directly connected to the policy questions raised in Chapter 1: whether to prioritise old cases, long-inactive cases, repeated-hearing cases, simpler cases, or case-type fairness. Third, the resulting model is transparent enough to support interpretable experiments rather than black-box optimisation.

Accordingly, the fundamental decision object in this chapter is a selected case set. The mathematical formulation will define the available pool, the selected subset, the capacity constraint, and the scoring objectives used to compare alternative policies.

2.3 Sets, Indices, and Basic Objects

Let $\mathcal{C}$ denote the complete curated pool of arbitration-related cases considered in the study. The present experiments focus on three case-type categories:

$$\mathcal{K} = \{\mathrm{FMAT}, \mathrm{AOCOM}, \mathrm{ADCOM}\}.$$

Each case $c \in \mathcal{C}$ has a case type $k_c \in \mathcal{K}$. At a scheduling step t, let $P_t \subseteq \mathcal{C}$ denote the cases available for selection. In a fully dynamic court model, P_t would change as new cases arrive, old cases are disposed, and listed matters return for further hearings. In the present experiments, P_t is treated as the available scheduling pool generated from the curated case dataset.

The scheduler selects a subset

$$S_t \subseteq P_t.$$

The size of the selected set is bounded by the listing capacity:

$$|S_t| \leq N_t,$$

where N_t represents the number of matters that can be selected at scheduling step t. In the experiments, N_t is treated as a fixed capacity parameter and is varied in capacity-sensitivity runs, for example using capacities such as 10, 20, and 30. This allows the thesis to study how scheduling behaviour changes when the system is more or less capacity-constrained.

For a selection-level formulation, it is useful to introduce a binary decision variable:

$$x_{c,t} = \begin{cases} 1, & \text{if case } c \text{ is selected at scheduling step } t, \\ 0, & \text{otherwise.} \end{cases}$$

Then the selected set can be written as:

$$S_t = \{c \in P_t : x_{c,t} = 1\},$$

and the capacity constraint becomes:

$$\sum_{c \in P_t} x_{c,t} \le N_t.$$

This is the core mathematical skeleton of the model. All scheduling policies studied in the thesis differ in how they assign priority to the candidate cases before selecting S_t.

2.4 Case-Level Features

Each case is represented by a small number of scheduling-relevant features. These features are not intended to capture the complete legal identity of a case. Instead, they capture the operational signals needed for the scheduling experiments.

For each case c, define:

$$A_c = \text{case age in days},$$

$$G_c = \text{gap since last hearing or order activity},$$

$$H_c = \text{hearing-history count},$$

$$O_c = \text{order count or activity-complexity proxy}.$$

The interpretation of these features is as follows. The age feature A_c measures how long a case has existed in the system. It represents the long-run backlog dimension of delay. The gap feature G_c measures inactivity: a case may not be extremely old, but it may still have remained untouched for a long period. The hearing-history feature H_c measures repeated engagement with the system. A large value of H_c may indicate that a case has consumed multiple listing opportunities without final closure. Finally, the order count O_c is used as a rough activity or complexity proxy. A case with many orders may be more procedurally involved than one with fewer recorded activities.

The case-type label k_c is also essential. In the present work, k_c belongs to FMAT, AOCOM, or ADCOM. The case type is used to study representation across categories. Without this feature, a scheduler could appear effective on delay metrics while repeatedly ignoring a smaller category.

The feature representation can therefore be written compactly as:

$$\phi(c) = (A_c, G_c, H_c, O_c, k_c).$$

A scheduling policy is a rule that maps the available feature vectors into a selected set:

$$\pi : \{\phi(c) : c \in P_t\} \to S_t.$$

This notation makes the modelling choice explicit: the policy is not using legal merits, confidential arguments, or judge-specific discretion. It is using measurable operational features to compare alternative scheduling philosophies.

2.5 Normalisation and Comparability of Features

The raw features have different scales. Case age and inactivity gap are measured in days, hearing history is a count, and order count is another count. Directly adding these quantities can make the largest-scale feature dominate the score. Therefore, for some experiments it is useful to work with normalised feature values.

For any numerical feature Z_c, define its min-max normalised value over the current pool P_t as:

$$\widetilde{Z}_c = \frac{Z_c - \min_{j \in P_t} Z_j}{\max_{j \in P_t} Z_j - \min_{j \in P_t} Z_j},$$

when the denominator is non-zero. If all cases have the same value of Z, the normalised value is taken as zero. This gives a scale-free version of the feature in the interval $[0, 1]$.

The single-objective policies can be understood directly in raw units because age and gap have meaningful interpretations. The bi-objective and multi-objective scalarisations use normalised versions where necessary, especially when delay and complexity are combined in the same score.

2.6 Single-Objective Scheduling Policies

The first experimental layer studies simple and interpretable single-objective policies. These policies are not intended to be final recommendations. They are baselines that isolate different scheduling intuitions.

2.6.1 Oldest-First or FCFS Policy

The oldest-first policy prioritises cases with the largest age:

$$\text{Score}_{FCFS}(c) = A_c.$$

The selected set is obtained by choosing the N_t cases with the largest values of A_c. This corresponds to the administrative intuition that the oldest matters should receive attention first. It is similar to a first-come-first-served rule when filing age is used as the ordering variable.

This policy is easy to explain and has strong intuitive appeal. Its limitation is that it ignores whether a case has recently moved. A very old case with recent activity may be prioritised over a moderately old case that has been inactive for a long time.

2.6.2 Gap-First Policy

The gap-first policy prioritises cases with the largest inactivity gap:

$$\text{Score}_{Gap}(c) = G_c.$$

This policy focuses on cases that have waited the longest since their last observed activity. It directly targets the experience of being left inactive. Its limitation is that it may select cases with long gaps even if their total age is not the largest.

2.6.3 Weighted Age-Gap Priority

The weighted age-gap policy combines age, inactivity gap, and hearing history:

$$D(c) = 0.5A_c + 0.4G_c + 0.1H_c.$$

The selected set is:

$$S_t = \arg \max_{S \subseteq P_t, |S| \leq N_t} \sum_{c \in S} D(c).$$

Because the objective is additive and there are no other constraints in this phase, the solution is obtained by sorting cases in decreasing order of $D(c)$ and selecting the top N_t cases.

This policy reflects the idea that delay should be measured through more than one signal. Age captures long-term pendency, gap captures inactivity, and hearing history captures repeated engagement with the court process.

2.6.4 Balanced Priority with Complexity Penalty

A slightly modified score introduces a mild penalty for order-count complexity:

$$B(c) = 0.45A_c + 0.35G_c + 0.15H_c - 0.05O_c.$$

The purpose of this score is to preserve delay priority while avoiding excessive emphasis on cases that may already have high procedural activity. The penalty is intentionally small because order count is only a proxy for complexity and should not dominate age or inactivity.

2.6.5 Shortest-Job-First Proxy

In classical scheduling, shortest-job-first prioritises tasks expected to finish quickly. In the court setting, true hearing duration or disposal probability is not directly modelled in this phase. Therefore, the thesis uses a proxy for complexity:

$$C(c) = H_c + 0.5O_c.$$

The SJF-style policy selects cases with low complexity:

$$\text{Score}_{SJF}(c) = -C(c).$$

This policy tests the opposite scheduling intuition from oldest-first selection. Instead of prioritising the most delayed matters, it asks what happens when the scheduler favours simpler matters. It may improve short-term manageability, but it risks leaving old and complex matters behind.

2.6.6 Round-Robin Case-Type Policy

The round-robin case-type policy is designed to improve representation across FMAT, AOCOM, and ADCOM. It maintains separate queues for each case type, sorts each queue by delay-relevant features, and then cyclically selects from the queues until the capacity is reached.

This is not a pure mathematical maximisation of age or gap. Instead, it encodes the principle that smaller case types should not be starved merely because another category dominates the pool. The policy is useful as a fairness baseline.

2.7 Coefficient Sensitivity

Weighted scores require coefficient choices. A score such as

$$\alpha A_c + \beta G_c + \gamma H_c - \delta O_c$$

is meaningful only if the coefficients reflect a scheduling philosophy. To avoid treating one coefficient setting as arbitrary, the thesis studies coefficient sensitivity.

The general weighted score is:

$$W(c; \alpha, \beta, \gamma, \delta) = \alpha A_c + \beta G_c + \gamma H_c - \delta O_c.$$

The profiles used in the experiments are:

Table 2.1: Coefficient profiles used for sensitivity analysis.

Profile	α	β	γ	δ	**Interpretation**
Age-dominant	0.70	0.20	0.10	0.00	Strong preference for old cases
Gap-dominant	0.30	0.60	0.10	0.00	Strong preference for long inactivity
History-dominant	0.40	0.20	0.40	0.00	Preference for repeat-hearing matters
Balanced	0.50	0.40	0.10	0.00	Main age-gap delay score
Complexity penalty	0.45	0.35	0.15	0.05	Delay priority with mild order-count penalty

The purpose of this analysis is not to find a perfect coefficient vector. It is to check whether the selected schedules change drastically when the emphasis shifts from age to gap to hearing history. If moderate changes produce similar selected sets, then the delay indicators are aligned in the dataset. If they produce very different sets, then coefficient choice becomes a major modelling decision.

2.8 Fairness as Case-Type Representation

Delay reduction is important, but a schedule that repeatedly selects only the dominant case category may be undesirable. In the present dataset, the relevant case types are FMAT, AOCOM, and ADCOM. To quantify representation, define the pool share of case type k as:

$$q_k(P_t) = \frac{|\{c \in P_t : k_c = k\}|}{|P_t|}.$$

Similarly, define the selected share of case type k as:

$$p_k(S_t) = \frac{|\{c \in S_t : k_c = k\}|}{|S_t|}.$$

The fairness error is defined as the L_1 distance between selected shares and pool shares:

$$E_{fair}(S_t) = \sum_{k \in \mathcal{K}} |p_k(S_t) - q_k(P_t)|.$$

A lower value of E_{fair} means that the selected set is closer to the case-type composition of the pool. For interpretability, the experiments also use a fairness score:

$$F(S_t) = 1 - \frac{1}{2} E_{fair}(S_t).$$

The division by 2 is used because the maximum possible L_1 distance between two probability distributions is 2. Thus, $F(S_t)$ lies between 0 and 1, with higher values indicating better case-type representation.

This fairness definition is deliberately modest. It does not claim to capture legal fairness in the full sense. It only measures whether the selected schedule preserves representation across the case types available in the dataset.

2.9 Bi-Objective Formulations

The next modelling step is to study pairs of objectives. The first bi-objective formulation compares delay priority with case-type fairness. The delay objective is:

$$D(S_t) = \frac{1}{|S_t|} \sum_{c \in S_t} D(c),$$

where

$$D(c) = 0.5A_c + 0.4G_c + 0.1H_c.$$

The fairness objective is $F(S_t)$ as defined above. The conceptual bi-objective problem is:

$$\max_{S_t \subseteq P_t} \big(D(S_t), F(S_t)\big) \quad \text{subject to} \quad |S_t| \leq N_t.$$

Since the two objectives may conflict, the experiments use a greedy scalarisation controlled by a fairness weight λ. When selecting cases one by one, the scheduler considers both the delay score of a candidate and the marginal fairness gain produced by adding that case. A simplified representation of the greedy score is:

$$\text{Score}_{DF}(c) = (1 - \lambda)\widetilde{D}(c) + \lambda \Delta F(c),$$

where $\widetilde{D}(c)$ is the normalised delay score and $\Delta F(c)$ is the improvement in fairness produced by adding case c to the partially selected set. When $\lambda = 0$, the scheduler behaves like a pure delay-priority scheduler. When $\lambda = 1$, it prioritises fairness gain.

The second bi-objective formulation compares delay priority with complexity control. Here the conceptual objective is:

$$\max D(S_t), \qquad \min C(S_t),$$

where

$$C(S_t) = \frac{1}{|S_t|} \sum_{c \in S_t} C(c).$$

This formulation captures the tension between selecting old or inactive cases and selecting matters that are likely to be simpler in terms of hearing and order history.

2.10 Multi-Objective Formulation

The final formulation combines the three main concerns of this thesis: delay priority, case-type fairness, and complexity control. The conceptual multi-objective problem is:

$$\max D(S_t), \qquad \max F(S_t), \qquad \min C(S_t),$$

subject to

$$S_t \subseteq P_t, \qquad |S_t| \leq N_t.$$

A fully general solution would search for a Pareto frontier of schedules. For the present thesis, the goal is more modest: to compare interpretable policy profiles. The experiments therefore use a greedy weighted scalarisation. At each selection step, a candidate case receives the score:

$$\text{Score}_{MO}(c) = w_D \widetilde{D}(c) + w_F \Delta F(c) - w_C \widetilde{C}(c),$$

where w_D, w_F, and w_C are non-negative weights satisfying:

$$w_D + w_F + w_C = 1.$$

The term $\widetilde{D}(c)$ rewards delay priority, $\Delta F(c)$ rewards marginal improvement in case-type representation, and $\widetilde{C}(c)$ penalises complexity. The scheduler repeatedly selects the candidate with the highest current score until the capacity is filled.

The policy profiles used in the experiments are listed in Table 2.2.

These profiles are intentionally simple. Their purpose is to show how the selected schedule changes as emphasis moves from delay to fairness to complexity. This is also why more complex methods such as genetic algorithms or reinforcement learning are left for future work. The present chapter first establishes clear, interpretable mathematical baselines.

Table 2.2: Multi-objective profiles used in the experiments.

Profile	w_D	w_F	w_C	Interpretation
Delay only	1.00	0.00	0.00	Pure delay-priority scheduling
Fairness only	0.00	1.00	0.00	Case-type representation dominates
Complexity only	0.00	0.00	1.00	Least-complex cases are preferred
Delay + fairness	0.60	0.30	0.10	Delay first, fairness second, mild complexity penalty
Delay + complexity	0.60	0.10	0.30	Delay first, with stronger complexity control
Fair-balanced	0.40	0.40	0.20	Balanced delay and fairness with moderate complexity penalty
Balanced all	0.50	0.30	0.20	Practical compromise across all three goals

2.11 Evaluation Metrics

The selected schedules are evaluated using metrics that correspond to the objectives and concerns described above.

For delay, the main metrics are:

$$\overline{A}(S_t) = \frac{1}{|S_t|} \sum_{c \in S_t} A_c,$$

$$\overline{G}(S_t) = \frac{1}{|S_t|} \sum_{c \in S_t} G_c.$$

These measure the average age and average inactivity gap of selected cases. A higher value indicates that the policy is selecting older or longer-inactive matters.

For hearing history and complexity, the metrics are:

$$\overline{H}(S_t) = \frac{1}{|S_t|} \sum_{c \in S_t} H_c,$$

$$\overline{C}(S_t) = \frac{1}{|S_t|} \sum_{c \in S_t} C_c.$$

For case-type representation, the selected counts and shares of FMAT, AOCOM, and ADCOM are reported. The fairness score $F(S_t)$ is used where a single summary number is needed.

For backlog impact, the experiments also report the number of old cases left after selection. If θ is the age threshold used to define an old case, then:

$$\mathrm{OldLeft}(P_t, S_t) = |\{c \in P_t \setminus S_t : A_c \geq \theta\}|.$$

In the experiments, the threshold is taken as one year, i.e., $\theta = 365$ days. This metric is important because a policy may select some old cases while still leaving many old cases behind.

2.12 Why These Approaches Are Suitable

The methods used in this thesis are intentionally simple and interpretable. At this stage, the goal is not to build a black-box scheduler, but to understand how different scheduling priorities change the selected set of cases. This is important because any computational support for judicial listing must remain explainable and auditable.

The single-objective policies isolate one administrative intuition at a time: oldest-first prioritises age, gap-first prioritises inactivity, SJF proxy prioritises simpler matters, and round-robin prioritises case-type representation. This makes the behaviour of each policy easy to explain and compare.

The coefficient-sensitivity experiment checks whether the weighted objective is overly dependent on one coefficient choice. The bi-objective experiments then expose the main trade-offs: improving fairness can reduce pure delay priority, while reducing complexity can leave older cases behind.

Finally, the multi-objective formulation combines the three practical goals that emerge from the earlier analysis: reducing delay, avoiding starvation of smaller case categories, and controlling schedule complexity. The greedy scalarisation used here is not claimed to be globally optimal; its value lies in being transparent, implementable, and directly connected to the feature-based abstraction. It also gives a reasonable baseline before moving to more advanced techniques such as genetic algorithms or reinforcement learning.

Thus, Chapter 2 provides the mathematical language for the experiments, while Chapter 3 applies it to the scheduling runs and results.

Chapter 3

Experiments, Results and Discussion

3.1 Experimental Overview

The previous chapter formulated the listing of arbitration matters as a capacity-constrained scheduling problem. This chapter applies that formulation to the curated Calcutta High Court arbitration-related dataset and studies how different scheduling policies behave under the same case pool and capacity assumptions.

The purpose of the experiments is not to claim that one rule can directly replace court discretion. Rather, the purpose is to make the trade-offs visible. A listing policy that prioritises the oldest matters may reduce age-based backlog, but it may also over-select one dominant case type. A policy that rotates across case types may improve representation, but it may select cases with lower delay priority. Similarly, a policy that favours simpler matters may be easier to execute, but it may leave older or more complex matters behind.

The experiments are therefore organised in a gradual sequence. First, single-objective policies are compared to isolate different scheduling intuitions. Second, coefficient sensitivity is tested to understand whether weighted priority scores are stable or arbitrary. Third, bi-objective experiments are used to study delay against fairness and delay against complexity. Finally, a multi-objective formulation combines delay priority, case-type fairness, and complexity control.

This progression mirrors the modelling philosophy of the thesis. The work begins with simple, interpretable policies before moving to combined objectives. Such an approach is useful because the judiciary scheduling problem is operationally sensitive:

any computational support system must be understandable, auditable, and capable of explaining why a particular set of matters was prioritised.

3.2 Dataset and Scraping Process

The dataset used in the experiments consists of arbitration-related matters from the Appellate Side of the Calcutta High Court. The focus is on cases scraped for the years 2024, 2025, and 2026. Within the available case records, three case-type categories were selected for scheduling experiments: FMAT, AOCOM, and ADCOM. These categories were chosen because, within the scraped Appellate Side data, they provided the closest usable operational pool for studying arbitration-related listing behaviour.

The data collection process was partly manual and partly script-assisted. Case details were collected from the Calcutta High Court/e-Courts interface, where case status, case history, filing and registration details, hearing dates, order information, and case metadata were available through case-detail pages. Since the public interface involves practical constraints such as manual access steps and CAPTCHA-like barriers, the scraping process required iterative manual supervision rather than being a fully automatic one-click pipeline.

The technical scraping details are not the main focus of this chapter. For the present experimental discussion, the important point is that the final processed dataset converts raw case information into scheduling features. Each case is represented through case age, last-activity gap, hearing-history count, order count, case type, and status-related information. These features are then used by the scheduling policies described in Chapter 2.

After cleaning and feature construction, the scheduling dataset contained 173 cases. Of these, 90 were FMAT matters, 74 were AOCOM matters, and 9 were ADCOM matters. Order-related information was available for most of the cases, with 168 cases having non-zero order information. Table 3.1 summarises the case-type composition of the experimental pool.

The imbalance in this table is important. ADCOM forms a small fraction of the available pool. Therefore, a purely delay-based scheduler may easily ignore ADCOM unless fairness is explicitly introduced. This is one of the reasons why the experiments later include round-robin, bi-objective, and multi-objective fairness-aware policies.

Table 3.1: Case-type composition of the experimental scheduling dataset.

Case Type	**Number of Cases**	**Approximate Share**
FMAT	90	52.0%
AOCOM	74	42.8%
ADCOM	9	5.2%
Total	**173**	**100.0%**

3.3 Experimental Setup

Each experiment is performed as a case-selection problem. The scheduler observes the available case pool and selects a limited number of cases according to a policy. The main capacity used in the baseline experiments is 20 cases. Additional capacity-sensitivity experiments use capacities of 10 and 30 cases to test whether the conclusions change when the listing constraint becomes tighter or more relaxed.

For each case c, the following features are used:

$$A_c = \text{case age in days},$$

$$G_c = \text{gap since last hearing or order activity},$$

$$H_c = \text{hearing-history count},$$

$$O_c = \text{order count or activity-complexity proxy}.$$

The case-type label is denoted by

$$k_c \in \{\text{FMAT}, \text{AOCOM}, \text{ADCOM}\}.$$

The main selected-set capacity constraint is:

$$|S| \leq N,$$

where S is the selected set of cases and N is the listing capacity.

The experiments compare the following scheduling policies:

1. **FCFS/Oldest-first:** selects cases with the largest case age.
2. **Gap-first:** selects cases with the largest inactivity gap.
3. **Weighted age-gap:** selects cases using a weighted delay score based on age, inactivity gap, and hearing history.
4. **Balanced priority:** uses a similar weighted delay score but includes a mild complexity penalty.
5. **SJF proxy:** selects cases with lower estimated complexity, using hearing-history count and order count as proxies.
6. **Round-robin case type:** rotates selection across FMAT, AOCOM, and ADCOM to improve representation.

The weighted delay-priority score used in the main single-objective experiment is:

$$D(c) = 0.5A_c + 0.4G_c + 0.1H_c.$$

The balanced priority score is:

$$B(c) = 0.45A_c + 0.35G_c + 0.15H_c - 0.05O_c.$$

The complexity proxy used for SJF-like and complexity-aware policies is:

$$C(c) = H_c + 0.5O_c.$$

The evaluation metrics are chosen to reflect the goals of the scheduler. The experiments report the average selected case age, average selected inactivity gap, average hearing-history count, old cases selected, case-type mix of the selected set, and old cases left in the backlog. In multi-day simulations, selected cases are removed from the immediate listing pool as a first approximation, and the remaining backlog is tracked across days. This is a simplified model of listing progress; richer stochastic execution is left for future work.

The rest of this chapter presents the results in the same order as the experimental design: single-objective scheduling, coefficient sensitivity, bi-objective scheduling, and multi-objective scheduling.

3.4 Overall Scheduling Outputs

Before analysing each experiment in detail, it is useful to summarise what the implemented scheduling algorithms produce at a high level. Each algorithm takes the same curated case pool as input and returns a selected set of cases subject to the listing capacity. Thus, the output of the experiment is not only a score or a plot, but an actual prioritised schedule of matters that would be listed under that policy.

In the baseline setting, the capacity is fixed at 20 cases. The delay-oriented policies, namely FCFS/oldest-first, gap-first, weighted age-gap, and balanced priority, all select old cases aggressively. Each of these policies selects 20 old cases out of 20 available slots. However, their selected case-type mix is concentrated in FMAT and AOCOM, with ADCOM absent from the selected set. This shows that delay-priority scheduling is effective at surfacing old matters, but it does not automatically protect smaller case categories.

The SJF proxy produces a different type of schedule. It selects simpler matters with much lower hearing-history count, but its selected cases are younger on average and it leaves more old backlog behind. This shows the cost of prioritising ease of processing. The round-robin policy, in contrast, produces the most balanced case-type schedule among the single-objective baselines, selecting FMAT, AOCOM, and ADCOM matters together. However, this comes with a reduction in average selected age.

The bi-objective and multi-objective algorithms refine this picture. In the delay-fairness setting, ADCOM appears only after the fairness weight becomes sufficiently high. This means that fairness cannot be treated as a minor adjustment to a delay score; it must be explicitly weighted if representation is required. In the delay-complexity setting, increasing the complexity penalty reduces the selected complexity, but it does not by itself solve ADCOM underrepresentation. Finally, the multi-objective scheduler combines delay, fairness, and complexity, showing that different weight profiles can produce different practical schedules, although several moderate profiles collapse to similar selected sets because the strongest age-gap cases dominate the pool.

Table 3.2 summarises the broad behaviour of the implemented scheduling approaches.

Table 3.2: Overall behaviour of the implemented scheduling approaches.

Approach	Primary Scheduling Intuition	Observed Behaviour
FCFS / oldest-first	Prioritise oldest matters	Selects the oldest cases and all selected cases are old, but excludes ADCOM.
Gap-first	Prioritise longest inactive matters	Selects cases with high inactivity gap, but still concentrates on FMAT and AOCOM.
Weighted age-gap	Combine age, gap, and hearing history	Produces a strong delay-oriented schedule similar to FCFS and gap-first.
Balanced priority	Delay priority with mild complexity penalty	Similar to weighted age-gap, showing that a mild order-count penalty does not change the top selected set.
SJF proxy	Prefer simpler matters	Selects lower-complexity cases, but with lower average age and weaker old-backlog targeting.
Round-robin case type	Improve case-type representation	Includes ADCOM and balances case types, but reduces average selected age.
Bi-objective delay–fairness	Balance delay priority and representation	ADCOM appears only when fairness weight is sufficiently high; moderate fairness weights remain delay-dominated.
Bi-objective delay–complexity	Balance delay priority and simplicity	Reduces selected complexity, but does not by itself improve ADCOM representation.
Multi-objective scheduling	Combine delay, fairness, and complexity	Exposes the practical trade-off among old-case priority, representation, and manageable schedules.

This overview provides the main result direction of the chapter. The following sections then examine each experimental layer in detail, beginning with the single-objective policies.

3.5 Single-Objective Scheduling Results

The first experimental layer studies single-objective scheduling policies. The aim of this stage is to understand the behaviour of simple and interpretable rules before moving to combined objectives. In each run, the scheduler observes the same case pool and selects a fixed number of cases. For the main baseline experiment, the listing capacity is fixed at $N = 20$.

Mathematically, each single-objective policy assigns a scalar score $s(c)$ to every case c in the available pool P. The scheduler then selects the top N cases according to that score:

$$S^\star = \arg \max_{S \subseteq P,\ |S| \leq N} \sum_{c \in S} s(c).$$

Since the objective is additive and the only constraint is the cardinality constraint $|S| \leq N$, this optimisation can be solved by sorting all cases in descending order of score and selecting the first N cases. Different policies correspond to different choices of the scoring function $s(c)$.

The features used in these policies are:

$$A_c = \text{case age in days},$$

$$G_c = \text{gap since last hearing or order activity},$$

$$H_c = \text{hearing-history count},$$

$$O_c = \text{order count or activity-complexity proxy}.$$

The following six policies are compared:

1. **FCFS / oldest-first:**
$$s(c) = A_c.$$
This policy gives priority to the oldest cases.

2. **Gap-first:**
$$s(c) = G_c.$$
This policy gives priority to cases that have remained inactive for the longest period.

3. **Weighted age-gap:**
$$s(c) = 0.5A_c + 0.4G_c + 0.1H_c.$$
This policy combines age, inactivity gap, and hearing history into a single delay-priority score.

4. **Balanced priority:**
$$s(c) = 0.45A_c + 0.35G_c + 0.15H_c - 0.05O_c.$$
This is similar to the weighted age-gap policy but introduces a mild penalty for order-count complexity.

5. **SJF proxy:**

$$C(c) = H_c + 0.5O_c, \qquad s(c) = -C(c).$$

This policy favours cases with lower estimated complexity. It is inspired by shortest-job-first scheduling, but since exact hearing duration is not available, hearing history and order count are used as proxies.

6. **Round-robin case type:** This policy does not optimise a single numerical delay score. Instead, it maintains separate queues for FMAT, AOCOM, and ADCOM, and cyclically selects from them to improve case-type representation.

The policies therefore represent different scheduling philosophies. FCFS asks what happens if the oldest matters are prioritised. Gap-first asks what happens if inactivity is prioritised. Weighted age-gap asks whether a combined delay score gives a stronger signal. Balanced priority asks whether a mild complexity penalty changes the selected set. SJF proxy asks what happens if simpler cases are preferred. Round-robin asks what happens if case-type representation is explicitly protected.

Table 3.3 summarises the Day-1 selected set for each policy. A case is counted as old if its age is at least 365 days.

Table 3.3: Day-1 comparison of single-objective scheduling policies with capacity 20.

Policy	Avg. Age	Avg. Gap	Avg. Hearings	Old Selected	FMAT	AOCOM	ADCOM
FCFS / oldest-first	1177.80	978.45	23.30	20	15	5	0
Gap-first	1134.00	1056.20	16.10	20	15	5	0
Weighted age-gap	1158.25	1047.55	22.30	20	15	5	0
Balanced priority	1158.25	1047.55	22.30	20	15	5	0
SJF proxy	565.65	585.25	1.00	16	7	13	0
Round-robin case type	992.65	867.15	21.90	20	7	7	6

The first important observation is that the delay-oriented policies all select old cases aggressively. FCFS, gap-first, weighted age-gap, and balanced priority each select 20 old cases out of the 20 available listing slots. This shows that the current pool contains a strong set of long-pending or long-inactive matters, and that these cases are immediately surfaced when age or gap is used as the main signal.

Figure 3.1 shows the average age of selected cases across the policies. FCFS selects the oldest cases on average because it directly optimises case age. Weighted age-gap and balanced priority remain close to FCFS, which means that adding gap and hearing history does not pull the selected set away from old matters. Gap-first selects slightly younger cases on average, but it selects cases with a higher average inactivity gap. This distinction is important: age and inactivity are related but not identical. A scheduling system that only tracks age may miss cases that have been inactive for long periods, while a system that only tracks gap may not always select the oldest matters.

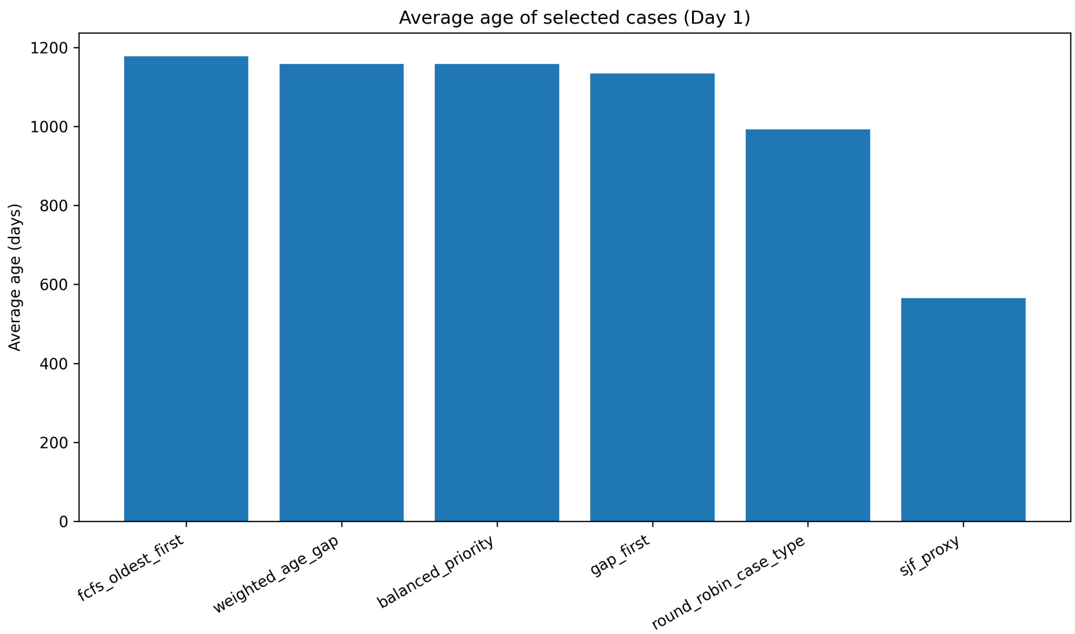

Figure 3.1: Average age of selected cases under single-objective scheduling policies.

The SJF proxy behaves differently. It selects cases with much lower average age and a very low average hearing-history count. This is expected because it is designed to favour cases that appear less complex. From a purely operational perspective, such a rule may be attractive because simpler matters may be easier to process. However, for backlog reduction, this policy is weaker: it selects fewer old cases and leaves more age-heavy backlog behind. This demonstrates the central limitation of a simplicity-first approach in a judicial context. A court may improve short-term manageability by selecting easier matters, but doing so can postpone difficult and old cases even further.

Figure 3.2 shows the selected case-type composition. The delay-oriented policies select 15 FMAT and 5 AOCOM cases, but no ADCOM cases. This is a significant result. It shows that a pure delay-priority scheduler can unintentionally ignore smaller case categories. ADCOM is already a minority category in the pool, and it does not appear among the highest-ranked delay-priority cases under these policies.

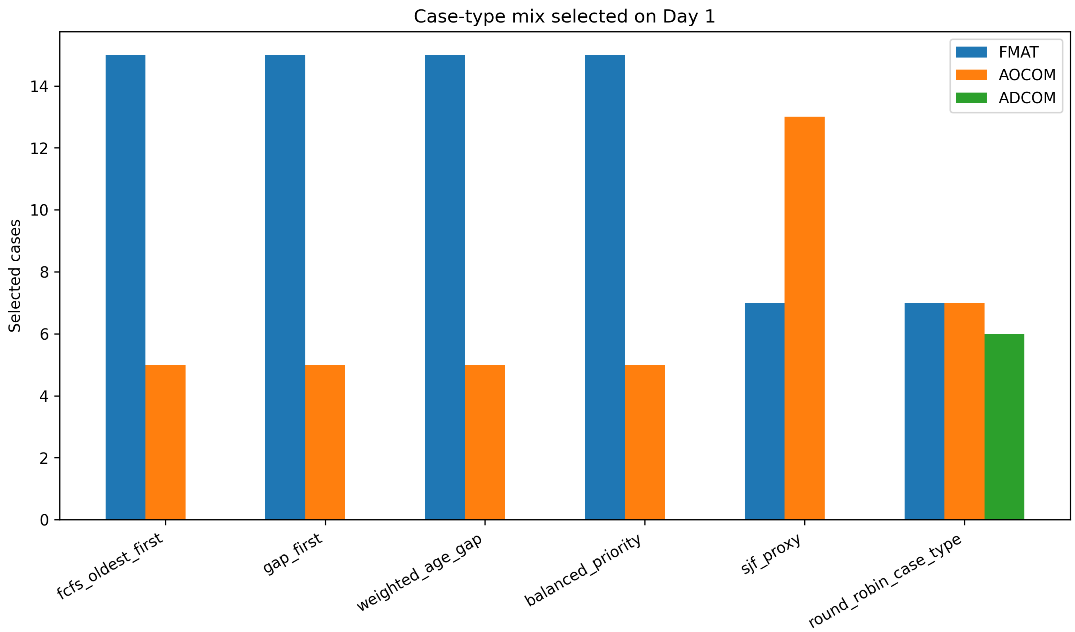

Figure 3.2: Case-type composition of selected cases under single-objective scheduling policies.

The round-robin policy directly corrects this representation issue. It selects 7 FMAT, 7 AOCOM, and 6 ADCOM cases. This is much more balanced than the delay-priority policies. However, this comes with a reduction in average selected age. The round-robin policy selects cases with an average age of 992.65 days, compared to 1177.80 days under FCFS and 1158.25 days under weighted age-gap. This is the first visible delay-fairness trade-off in the experiments: representation improves, but pure delay priority falls.

The weighted age-gap and balanced-priority policies produce the same aggregate selected set. This should not be read as a failure of the balanced score. Instead, it indicates that the penalty term $-0.05O_c$ is too mild to change the top-ranked cases in this dataset. In other words, the strongest old and inactive cases remain dominant even after a small complexity adjustment. This observation motivates the coefficient-sensitivity analysis in the next section.

Figure 3.3 shows the number of old cases left in the backlog across simulation days. In this simplified multi-day simulation, selected cases are removed from the immediate listing pool and the remaining cases continue to age. This does not claim to reproduce the full legal outcome of a hearing. Rather, it gives a controlled way to compare how quickly different policies remove old matters from the immediate scheduling backlog.

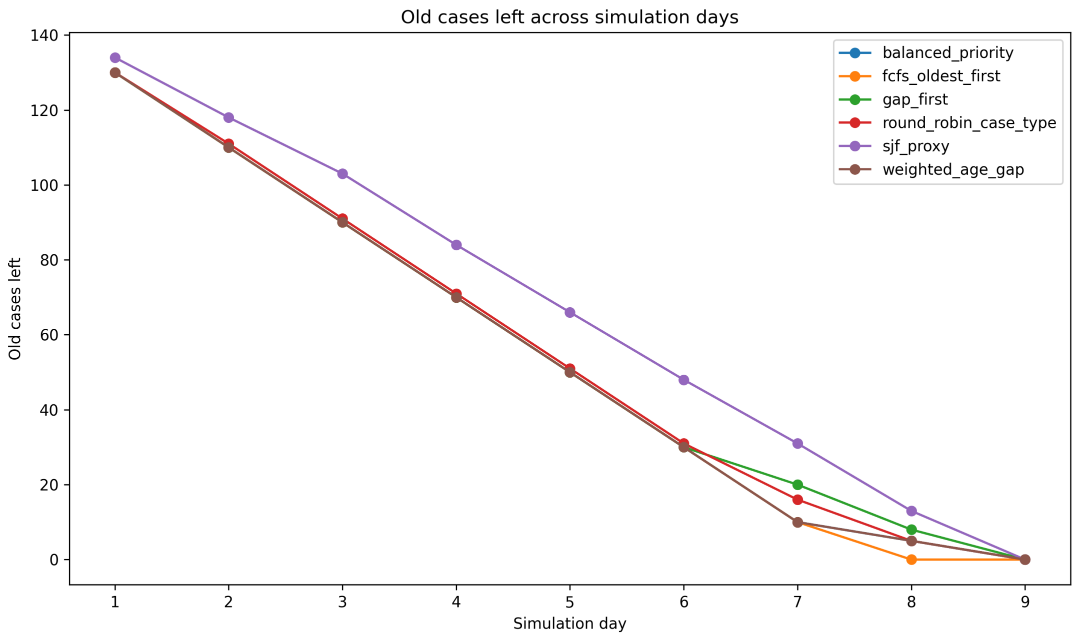

Figure 3.3: Old cases left in the backlog across simulation days under different single-objective policies.

The overall conclusion from the single-objective experiment is that no single policy dominates on all dimensions. FCFS is strongest for age. Gap-first is strongest for inactivity. Weighted age-gap and balanced priority provide robust delay-oriented selections. SJF proxy selects simpler matters but performs poorly for old-backlog reduction. Round-robin improves representation but sacrifices some delay priority. This motivates the next question: can the weighted scores be tuned to recover a better compromise, or is an explicit multi-objective formulation required?

3.6 Coefficient Sensitivity Analysis

The weighted policies depend on the choice of coefficients. For example, the weighted age-gap score

$$D(c) = 0.5A_c + 0.4G_c + 0.1H_c$$

implicitly states that age is slightly more important than inactivity gap, while hearing history is a smaller supporting signal. Similarly, the balanced score

$$B(c) = 0.45A_c + 0.35G_c + 0.15H_c - 0.05O_c$$

assigns a small penalty to order-count complexity. These choices are reasonable, but they should not be treated as self-evident. Therefore, a coefficient-sensitivity experiment is performed.

The general scoring function is:

$$W(c; \alpha, \beta, \gamma, \delta) = \alpha A_c + \beta G_c + \gamma H_c - \delta O_c.$$

Here, α controls the importance of case age, β controls the importance of inactivity gap, γ controls the importance of hearing-history count, and δ controls the strength of the order-count complexity penalty. Five coefficient profiles are compared:

Table 3.4: Coefficient profiles used in the sensitivity experiment.

Profile	α	β	γ	δ	**Interpretation**
Age-dominant	0.70	0.20	0.10	0.00	Strong old-case priority
Gap-dominant	0.30	0.60	0.10	0.00	Long-inactivity priority
History-dominant	0.40	0.20	0.40	0.00	Repeat-hearing priority
Balanced	0.50	0.40	0.10	0.00	Main delay-priority profile
Complexity penalty	0.45	0.35	0.15	0.05	Delay priority with mild complexity penalty

The purpose of this experiment is to answer two questions. First, does the selected set change substantially when the score places more emphasis on age, gap, or hearing history? Second, can a mild complexity penalty alter the selected case mix or reduce the tendency to select highly active cases?

Table 3.5 summarises the results. All coefficient profiles select 20 old cases. Moreover, all five profiles select the same case-type composition: 15 FMAT, 5 AOCOM, and 0 ADCOM. This shows that moderate coefficient changes do not meaningfully change the selected case-type mix.

Table 3.5: Coefficient sensitivity results for capacity 20.

Profile	Avg. Age	Avg. Gap	Avg. Hearings	Old Selected	FMAT	AOCOM	ADCOM
Age-dominant	1167.80	1033.60	24.25	20	15	5	0
Gap-dominant	1151.85	1052.05	20.15	20	15	5	0
History-dominant	1167.80	1033.60	24.25	20	15	5	0
Balanced	1158.25	1047.55	22.30	20	15	5	0
Complexity penalty	1158.25	1047.55	22.30	20	15	5	0

At first glance, this may appear uninteresting because the selected case mix does not change. However, it is actually a useful result. It indicates that the top-priority cases in the current pool are old, inactive, and often hearing-heavy at the same time. Therefore, age, gap, and hearing history are aligned among the strongest candidates. As a result, changing the coefficients within a moderate range does not radically alter the selected schedule.

Figure 3.4 shows the age-gap trade-off across the coefficient profiles. The profiles lie close to one another, confirming that the weighted objective is stable. The gap-dominant profile increases the average selected gap slightly, while the age-dominant and history-dominant profiles select slightly older and more hearing-heavy cases. However, the differences are small relative to the overall magnitude of the age and gap values.

Figure 3.5 shows the selected case-type shares under each coefficient profile. The plot confirms that coefficient tuning alone does not bring ADCOM into the selected set. This is important because it shows that fairness cannot be assumed to emerge automatically from a weighted delay score. If representation across FMAT, AOCOM, and ADCOM is desired, it must be introduced explicitly.

The coefficient-sensitivity experiment therefore has two interpretations. On the positive side, the weighted delay objective is robust: it does not depend excessively on one exact coefficient setting. On the limiting side, robustness also means that moderate coefficient changes are not enough to solve case-type underrepresentation. This leads naturally to the bi-objective stage, where fairness is introduced as an explicit objective rather than being left as an indirect consequence of score weighting.

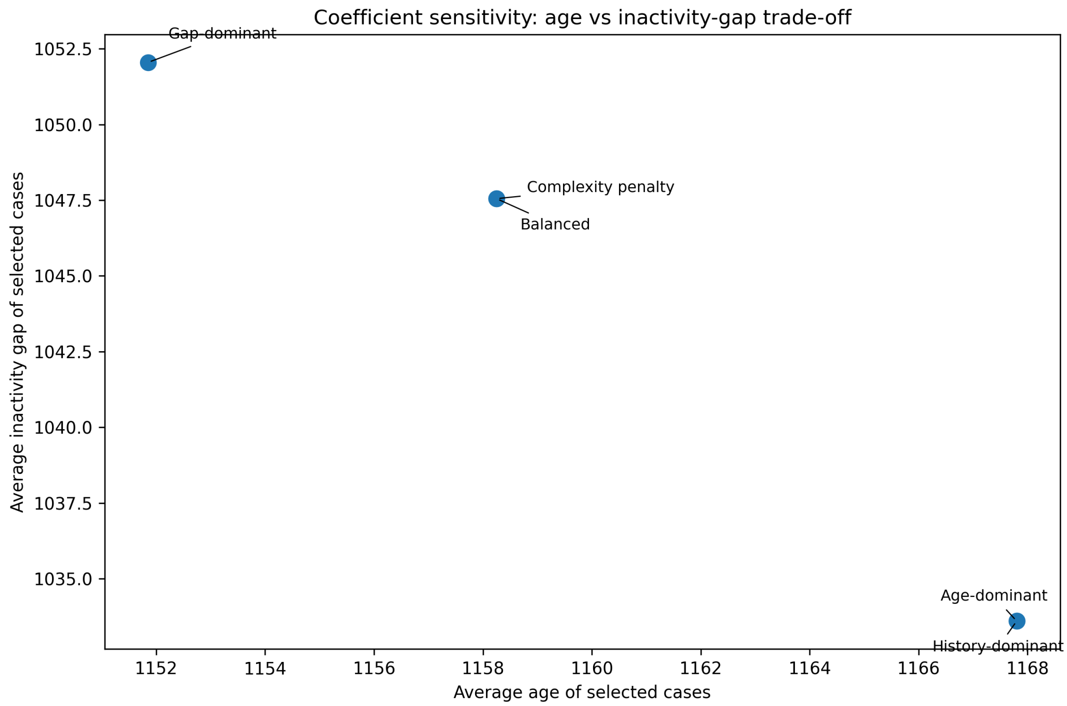

Figure 3.4: Coefficient sensitivity: trade-off between average selected age and average selected inactivity gap.

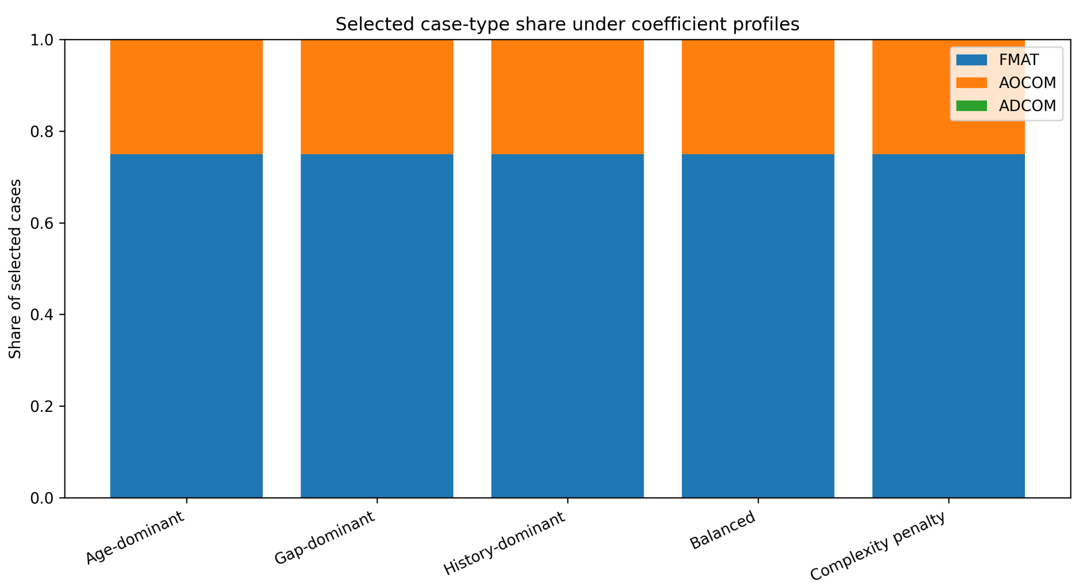

Figure 3.5: Selected case-type share under different coefficient profiles.

3.7 Bi-Objective Scheduling Results

The single-objective experiments showed that delay-oriented policies and fairness-oriented policies behave differently. Delay-based policies selected old and inactive matters, but they ignored ADCOM cases. Round-robin improved representation, but reduced the average selected age. The next step is therefore to study these objectives jointly rather than separately.

Two bi-objective experiments are considered. The first studies the trade-off between delay priority and case-type fairness. The second studies the trade-off between delay priority and complexity control. These two experiments answer different questions. The delay-fairness experiment asks whether a schedule can remain delay-aware while also representing FMAT, AOCOM, and ADCOM. The delay-complexity experiment asks whether the scheduler can continue selecting delayed cases while avoiding excessively complex matters.

3.7.1 Delay Priority versus Case-Type Fairness

The first bi-objective experiment combines the delay-priority score

$$D(c) = 0.5A_c + 0.4G_c + 0.1H_c$$

with the case-type fairness objective defined earlier. The fairness score measures how close the selected FMAT, AOCOM, and ADCOM shares are to their shares in the available pool. A higher fairness score indicates better representation.

The scheduler uses a fairness weight λ. When $\lambda = 0$, the policy behaves like a pure delay-priority scheduler. When $\lambda = 1$, the policy gives maximum importance to fairness gain. Intermediate values of λ show how the selected schedule changes as fairness is gradually introduced.

Table 3.6 summarises the results for capacity 20.

Table 3.6: Bi-objective delay-fairness results for capacity 20.

λ	**Delay Mean**	**Fairness Score**	**Avg. Age**	**Avg. Gap**	**FMAT**	**AOCOM**	**ADCOM**
0.00	1000.38	0.770	1158.25	1047.55	15	5	0
0.10	1000.38	0.770	1158.25	1047.55	15	5	0
0.25	1000.38	0.770	1158.25	1047.55	15	5	0
0.50	1000.38	0.770	1158.25	1047.55	15	5	0
0.75	943.05	0.970	1086.70	994.60	11	8	1
0.90	943.05	0.970	1086.70	994.60	11	8	1
1.00	625.21	0.978	753.80	616.45	10	9	1

The main observation is that low fairness weights do not change the selected schedule. For $\lambda = 0$ to $\lambda = 0.50$, the selected set remains identical: 15 FMAT cases, 5 AOCOM cases, and no ADCOM cases. This indicates that the delay-priority signal is strong enough to dominate moderate fairness weights. In practical terms, merely adding a small fairness term is not sufficient to correct underrepresentation.

A clear change occurs at $\lambda = 0.75$. At this point, the selected set includes ADCOM for the first time, and the case-type distribution becomes 11 FMAT, 8 AOCOM, and 1 ADCOM. The fairness score improves sharply from 0.770 to 0.970, while the delay-priority mean falls from 1000.38 to 943.05. This is a meaningful compromise: the schedule becomes much more representative, but it still retains a large part of the delay-priority benefit.

At $\lambda = 1.00$, fairness improves only slightly further, from 0.970 to 0.978, but the delay-priority score drops sharply to 625.21. The average selected age also falls to 753.80 days. This suggests that pure fairness is not the best practical choice if delay reduction remains important. The best compromise in this experiment lies around $\lambda = 0.75$ or $\lambda = 0.90$, where ADCOM is included and fairness improves substantially without the full delay sacrifice of the pure fairness setting.

Figure 3.6 shows the delay-fairness trade-off. The overlapping low-λ points are not merely a plotting issue. They show that the same selected schedule remains stable for a range of fairness weights. This stability is useful because it reveals a threshold effect: fairness must be weighted strongly enough before the selected set changes.

Figure 3.7 shows the selected case-type composition as λ increases. The plot makes the transition visible: ADCOM is absent when fairness is weak, appears when fairness becomes strong enough, and remains present under pure fairness. This confirms that case-type representation must be modelled explicitly rather than expected to emerge automatically from delay-based scores.

The delay-fairness experiment therefore gives an important scheduling insight. If the objective is only delay reduction, smaller case categories can be starved. If the objective is only representation, the scheduler may select substantially younger or lower-delay cases. A mixed policy with sufficiently high but not absolute fairness weight provides a more balanced schedule.

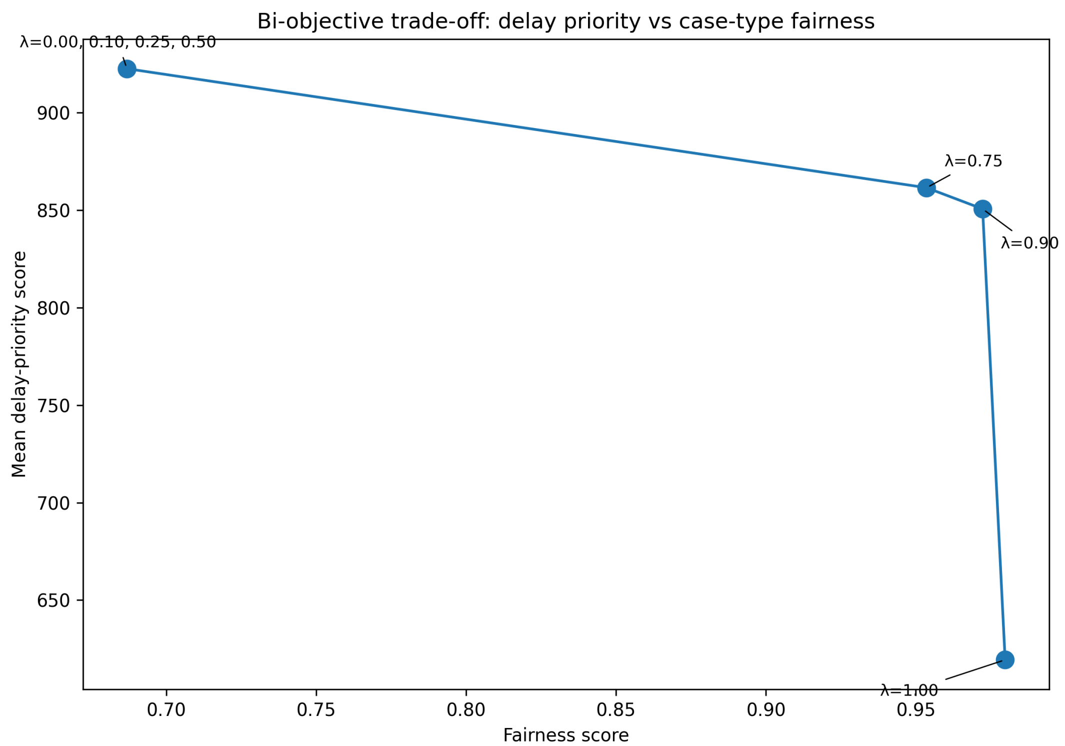

Figure 3.6: Bi-objective frontier between delay priority and case-type fairness.

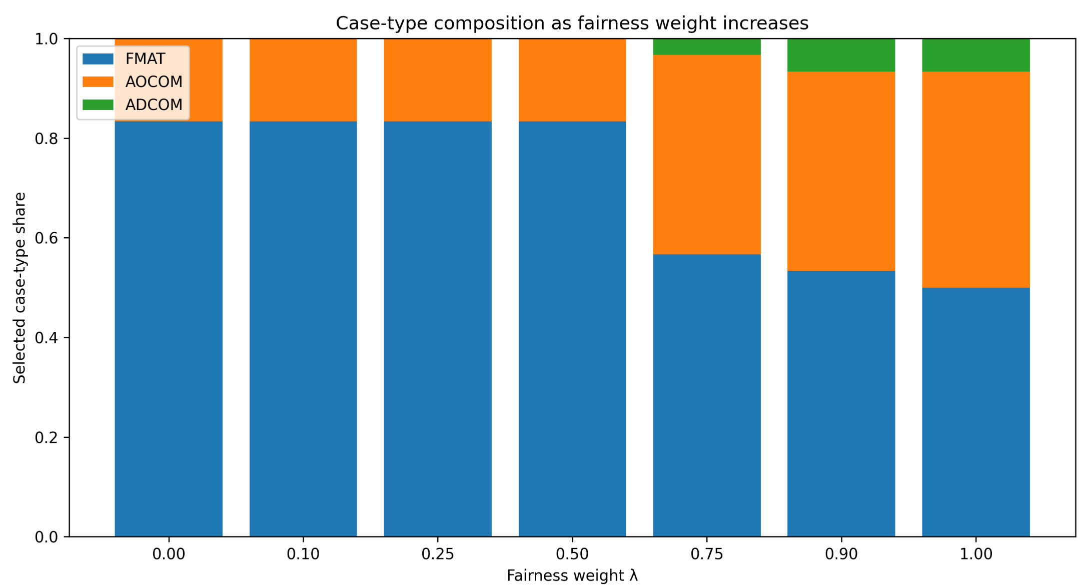

Figure 3.7: Selected case-type composition as fairness weight increases.

3.7.2 Delay Priority versus Complexity

The second bi-objective experiment studies delay priority against complexity control. The motivation is different from the fairness experiment. A delay-oriented scheduler may select old cases that have already accumulated many hearings and orders. Such cases may be important, but they may also be difficult to move quickly. A complexity-aware scheduler can reduce this burden, but it may sacrifice delay priority.

The complexity score is defined as:

$$C(c) = H_c + 0.5 O_c.$$

A complexity penalty weight λ is used. When $\lambda = 0$, the scheduler is pure delay-priority. When $\lambda = 1$, the scheduler strongly favours low-complexity matters.

Table 3.7 summarises the results.

Table 3.7: Bi-objective delay-complexity results for capacity 20.

λ	**Delay Mean**	**Complexity Mean**	**Avg. Age**	**Old Selected**	**FMAT**	**AOCOM**	**ADCOM**
0.00	1000.38	23.53	1158.25	20	15	5	0
0.10	992.98	13.33	1145.00	20	15	5	0
0.25	981.24	9.55	1129.30	20	16	4	0
0.50	969.67	8.13	1105.30	20	16	4	0
0.75	901.14	3.68	1036.55	20	16	4	0
0.90	835.47	2.50	947.00	20	15	5	0
1.00	517.03	1.45	565.65	16	7	13	0

The table shows a smooth trade-off. As the complexity penalty increases, the mean selected complexity falls from 23.53 to 1.45. However, the delay-priority score also falls, and the average selected age drops from 1158.25 days to 565.65 days under pure complexity minimisation. This confirms that reducing complexity can make the selected schedule simpler, but the cost is weaker delay targeting.

A second important observation is that ADCOM does not appear in any delay-complexity setting. Even when the scheduler strongly favours low-complexity cases, ADCOM is still not selected. This shows that complexity control is not a substitute for fairness. If ADCOM representation is required, it must be introduced through a fairness objective or a case-type-aware rule.

Figure 3.8 shows the delay-complexity frontier. The curve illustrates the cost of reducing complexity: as selected matters become simpler, the delay-priority score decreases. This figure complements the earlier fairness frontier. Together, the two bi-objective experiments show that there are two distinct trade-offs: delay versus representation, and delay versus manageability.

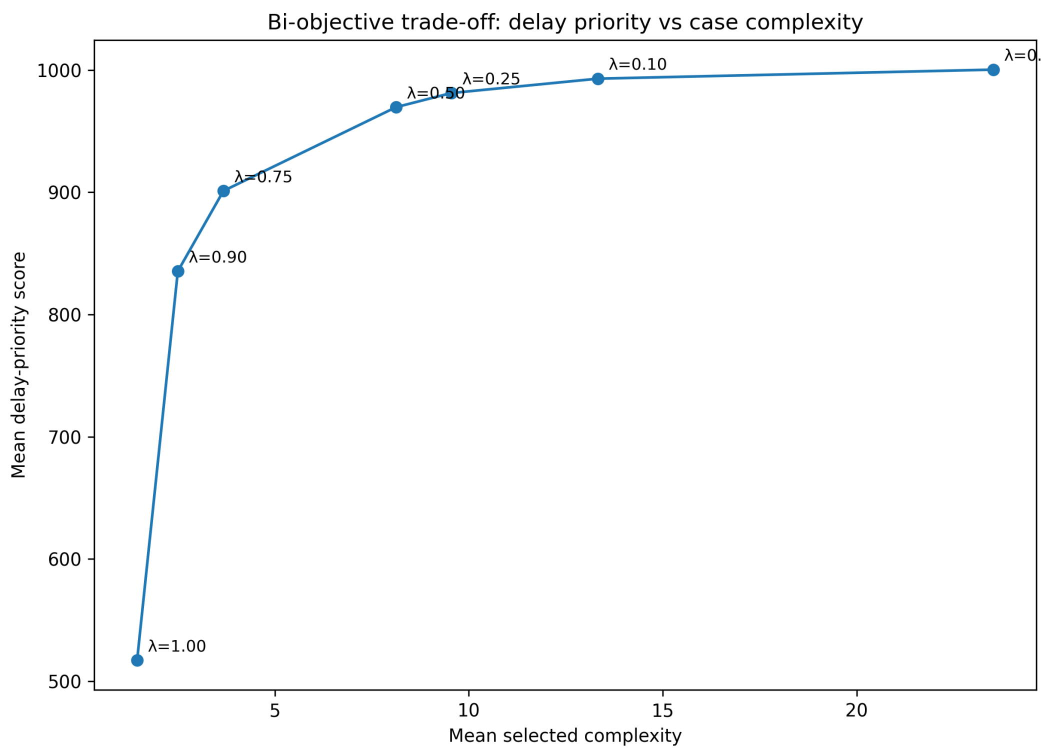

Figure 3.8: Bi-objective trade-off between delay priority and selected-case complexity.

The bi-objective experiments make it clear that the scheduling problem cannot be reduced to a single score without losing important information. Fairness and complexity affect the selected set in different ways. This motivates the final experimental layer, where delay priority, fairness, and complexity are combined into a single multi-objective scheduling framework.

3.8 Multi-Objective Scheduling Results

The final experimental layer combines three objectives: delay priority, case-type fairness, and complexity control. This is closest to the practical structure of court scheduling. A useful schedule should not ignore old or inactive cases, should not starve smaller case categories, and should not become dominated by highly complex matters.

The multi-objective score used in the experiment is:

$$\text{Score}(c) = w_D \widetilde{D}(c) + w_F \Delta F(c) - w_C \widetilde{C}(c),$$

where w_D is the delay weight, w_F is the fairness weight, and w_C is the complexity penalty weight. The weights satisfy:

$$w_D + w_F + w_C = 1.$$

The scheduler selects cases greedily. At each step, it chooses the case with the highest current multi-objective score, where the fairness term is computed as the marginal improvement in case-type representation after adding that case.

Table 3.8 summarises the multi-objective results for capacity 20.

Table 3.8: Multi-objective scheduling results for capacity 20.

Profile	Delay	Fairness	Complexity	Avg. Age	FMAT	AOCOM	ADCOM	Old
Delay only	1000.38	0.770	23.53	1158.25	15	5	0	20
Fairness only	625.21	0.978	19.05	753.80	10	9	1	20
Complexity only	555.75	0.978	1.45	611.15	10	9	1	20
Delay + fairness	992.98	0.770	13.33	1145.00	15	5	0	20
Delay + complexity	981.24	0.720	9.55	1129.30	16	4	0	20
Fair-balanced	981.24	0.720	9.55	1129.30	16	4	0	20
Balanced all	981.24	0.720	9.55	1129.30	16	4	0	20

The delay-only profile gives the highest delay-priority score, but it selects no ADCOM cases. This is consistent with the earlier single-objective and bi-objective results. Pure delay optimisation repeatedly favours FMAT and AOCOM because the strongest age-gap signals are concentrated there.

The fairness-only profile gives the best case-type representation. It selects 10 FMAT, 9 AOCOM, and 1 ADCOM case, with a fairness score of 0.978. However, the delay-priority score falls to 625.21. This is the same trade-off seen in the bi-objective experiment: fairness improves representation, but the selected set becomes less delay-heavy.

The complexity-only profile minimises selected-case complexity most strongly. Its mean complexity is only 1.45. Interestingly, it also achieves a high fairness score and includes ADCOM. However, it has the lowest delay-priority score among the main profiles. This confirms that low complexity does not necessarily correspond to high delay priority. A schedule that is easy to process may not be the schedule that best addresses old backlog.

The delay-fairness profile preserves most of the delay benefit while reducing complexity substantially compared to the delay-only profile. Its delay score remains high at 992.98, while its complexity falls from 23.53 to 13.33. However, it still selects no ADCOM cases. This shows that a moderate fairness term is not always enough to change the selected case-type composition when the delay signal is strong.

The profiles delay-complexity, fair-balanced, and balanced-all all converge to the same selected schedule. This is an important result. It means that different moderate weight combinations can collapse to the same practical selection because the top-ranked cases remain similar in the combined objective space. In the current dataset, the age-gap signal is strong enough that moderate fairness and complexity terms do not fully overturn it. The overlapping points in the plots should therefore be interpreted as a modelling insight, not merely as a visual artefact.

Figure 3.9 shows the multi-objective schedules in the delay-fairness plane. It illustrates the same structure: delay-only and delay-heavy mixed profiles remain high on delay but weaker on representation, while fairness-only moves toward better representation with lower delay priority.

Figure 3.10 shows the profiles in the delay-complexity plane. The figure highlights the role of the complexity penalty. Delay-only selects high-delay but also high-complexity matters. Complexity-only selects very simple matters but sacrifices delay. The mixed profiles lie between these extremes.

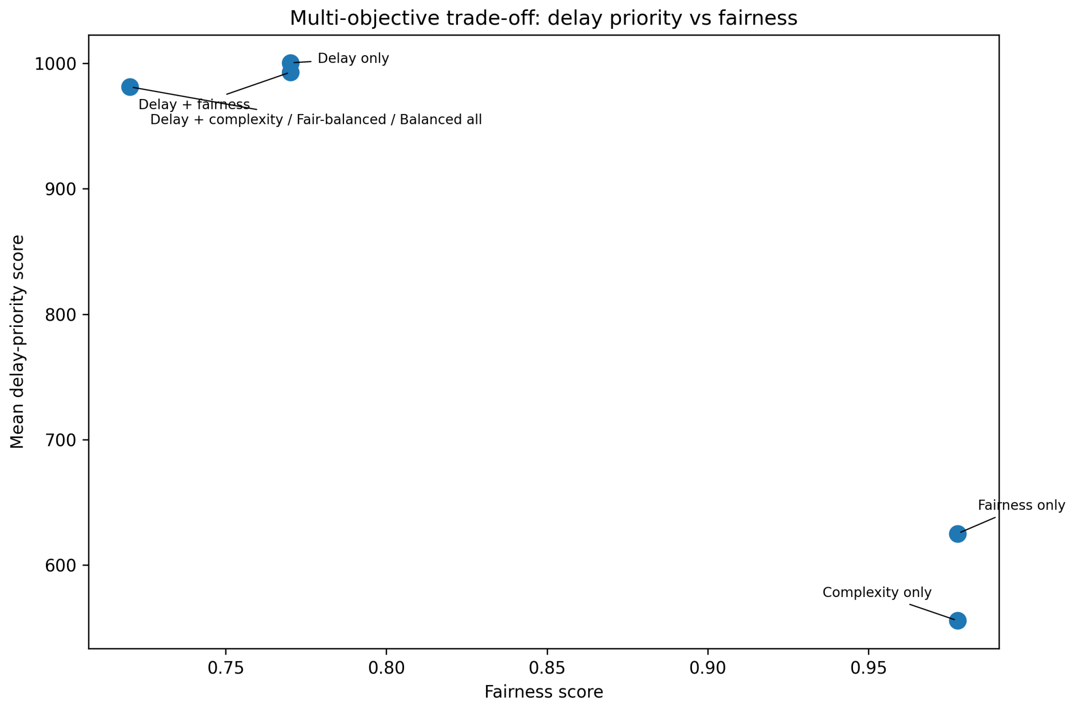

Figure 3.9: Multi-objective profiles plotted in the delay-fairness objective space.

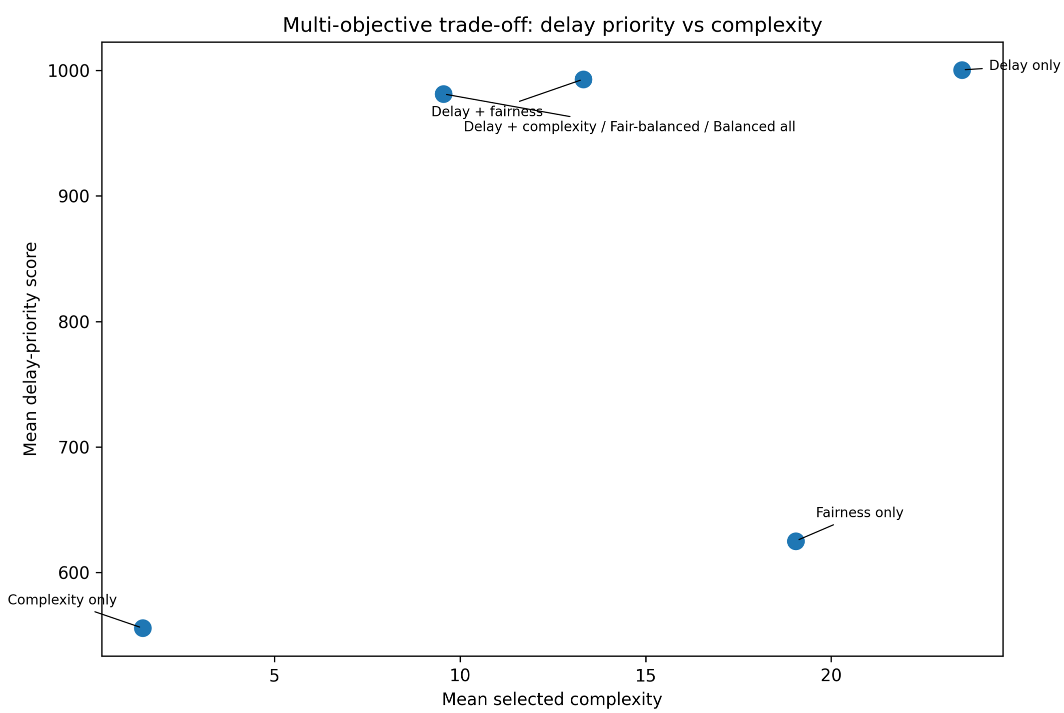

Figure 3.10: Multi-objective profiles plotted in the delay-complexity objective space.

Figure 3.11 shows the selected case-type composition under the multi-objective profiles. The plot again confirms that ADCOM appears only when fairness or simplicity becomes dominant enough. Delay-heavy mixed profiles continue to exclude ADCOM. This supports a central finding of the thesis: smaller case-type representation cannot be treated as a side effect of delay optimisation; it must be explicitly encoded.

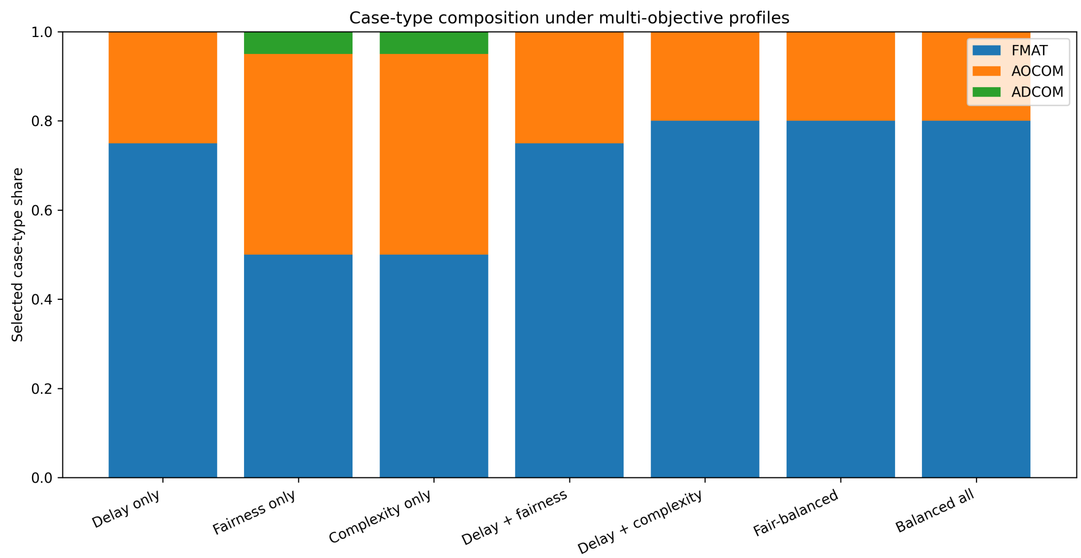

Figure 3.11: Case-type composition under multi-objective scheduling profiles.

3.8.1 Capacity Sensitivity in the Multi-Objective Setting

The final robustness check studies capacity sensitivity. The same multi-objective profiles are run at capacities 10, 20, and 30. Capacity matters because the scheduling trade-off becomes sharper when fewer cases can be listed. Under low capacity, the scheduler must choose among many old and inactive matters, so every selected slot is more consequential. Under higher capacity, there is more room to include additional categories, but delay-heavy policies may still over-select the dominant case type if fairness is not explicitly prioritised.

Figure 3.12 plots the delay-fairness space under different capacities. The plot shows that capacity changes the location of the profiles, but the qualitative trade-off remains. Delay-oriented profiles stay high on delay and weaker on fairness. Fairness-oriented profiles move closer to the pool-level case-type distribution but lose delay quality. This indicates that the trade-off is structural rather than an artefact of a single capacity setting.

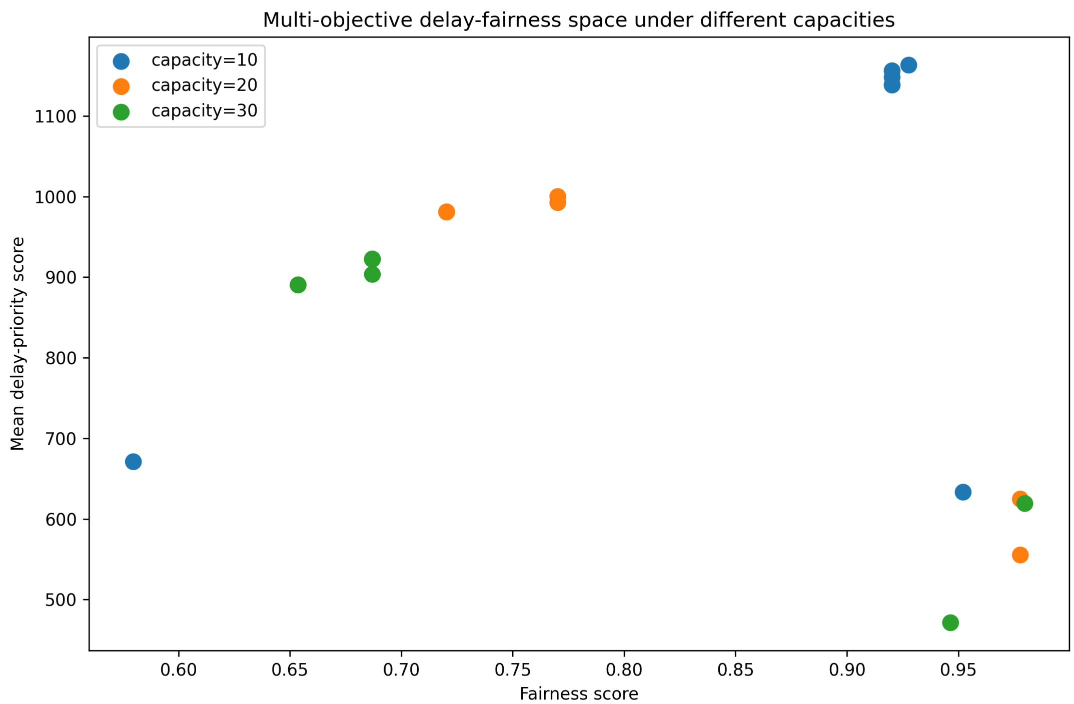

Figure 3.12: Multi-objective delay-fairness space under different listing capacities.

The capacity experiment also shows why fixed-capacity results should be interpreted carefully. A policy that appears sufficiently fair at capacity 10 may become less representative at capacity 30 if the additional slots are filled mostly by the dominant case type. Conversely, a fairness-aware policy may be more valuable when the capacity is large enough to allow representation without entirely sacrificing delay. Thus, capacity is not only a parameter; it affects the practical meaning of each objective.

Overall, the multi-objective results show that an integrated formulation is necessary for this problem. Single-objective policies are useful for interpretation, but they cannot capture all scheduling concerns simultaneously. Bi-objective policies expose pairwise trade-offs, but real listing decisions involve more than two goals. The multi-objective experiment combines these goals and shows how the selected schedule changes as the relative importance of delay, fairness, and complexity changes.

3.9 Overall Discussion and Key Findings

The experiments in this chapter show that the scheduling formulation developed in Chapter 2 produces meaningful and interpretable outputs. Each policy generates a concrete selected set of cases, and the differences between these selected sets reveal the practical consequences of choosing one scheduling philosophy over another.

The first major finding is that delay-oriented policies are effective at surfacing old matters. FCFS/oldest-first, gap-first, weighted age-gap, and balanced-priority policies all selected old cases aggressively. This confirms that simple feature-based rules can identify cases that deserve attention from a delay-reduction perspective. However, the results also show that age and inactivity are not identical. FCFS selects the oldest matters, while gap-first selects cases that have waited longest since last activity. A good court-listing policy should therefore not treat delay as a single-dimensional quantity.

The second finding is that the strongest delay signals in the dataset are concentrated in FMAT and AOCOM matters. Across the single-objective delay-based policies, ADCOM cases were not selected in the top-20 schedule. This is not because the scheduler failed technically; rather, it reveals a structural issue in the data and objective design. If a smaller case category does not appear among the highest age-gap cases, a pure delay-priority rule will ignore it. Thus, representational fairness cannot be expected to emerge automatically from delay optimisation.

The third finding is that coefficient tuning alone is not sufficient to fix this issue. The coefficient-sensitivity experiment showed that moderate changes in the weights of age, inactivity gap, hearing history, and order count produced very similar selected sets. This is useful because it suggests that the weighted delay objective is stable. However, it also means that if the policy objective is fundamentally delay-heavy, minor coefficient adjustments will not substantially alter the case-type composition. Fairness must be introduced explicitly if case-type representation is desired.

The fourth finding is that fairness has a visible cost, but the cost is not linear. In the bi-objective delay-fairness experiment, low fairness weights produced no change in the selected schedule. Once the fairness weight became sufficiently large, ADCOM entered the selected set and the fairness score improved sharply. However, a pure fairness setting reduced the delay-priority score substantially. This suggests that a moderate-to-high fairness weight can be practically useful, while pure fairness may overcorrect if delay reduction remains a major goal.

The fifth finding is that complexity control addresses a different concern from fairness. The delay-complexity experiment showed that increasing the complexity penalty reduces the selected complexity of the schedule. However, complexity control alone did not bring ADCOM into the selected set. This shows that simplicity and representation are distinct objectives. A schedule can be simpler without being more representative, and a representative schedule need not be the least complex one.

The final multi-objective experiment combines these insights. Delay, fairness, and complexity are all relevant, but they pull the scheduler in different directions. The delay-only profile gives the strongest delay-priority score but weak case-type representation. The fairness-only profile improves representation but lowers delay quality. The complexity-only profile selects simpler matters but weakens backlog targeting. Mixed profiles provide intermediate behaviour, and in some cases different weight combinations converge to the same selected schedule. This convergence itself is informative: in the current dataset, the strongest age-gap cases remain dominant unless fairness or complexity is weighted strongly enough to change the selected set.

Overall, the experiments demonstrate that court-listing policies should be evaluated as multi-objective decision rules. A policy cannot be judged only by how old the selected cases are, nor only by how balanced the case types are, nor only by how simple the selected matters appear. Each of these dimensions captures one part of the scheduling problem. The value of the proposed framework is that it makes these dimensions measurable and comparable.

3.10 Limitations of the Experimental Setup

The experiments in this chapter are intentionally limited in scope. The first limitation is that the model operates at the selection level. It decides which cases should be prioritised for listing, but it does not simulate the exact order in which matters are heard during the court day, or whether a listed case is actually heard, adjourned, passed over, or disposed.

The second limitation is that the current model does not explicitly include judge availability, advocate conflicts, bench-specific roster constraints, or stakeholder absence. These factors were identified as important in the empirical motivation, but they require richer operational data and a more detailed simulator, and are therefore treated as future extensions.

The third limitation is that hearing duration and disposal probability are not predicted. Since such information is not fully modelled here, the SJF proxy uses hearing-history count and order count as approximate complexity signals, which is useful for comparison but not a substitute for a true duration or disposal model.

The fourth limitation is that fairness is defined only in terms of case-type representation across FMAT, AOCOM, and ADCOM. This is a reasonable first scheduling-fairness metric, but a more complete model may include fairness across age buckets, litigant categories, benches, advocates, urgency classes, or statutory sections.

Despite these limitations, the experimental setup provides a transparent baseline. It converts a broad institutional problem into a measurable scheduling task and shows how different scheduling objectives behave and where their trade-offs arise.

3.11 What Next

The results of this chapter suggest that the present formulation is a useful first step, but not the final form of the scheduling system. The current implementation uses interpretable greedy and rule-based policies, which are appropriate for establishing baselines, but richer versions of the problem will involve more constraints, more uncertainty, and a larger search space.

The next chapter discusses these extensions: a more realistic stochastic simulator where listed cases may be heard, adjourned, not reached, disposed, or returned to the queue; richer constraints such as judge and advocate availability, bench compatibility, hearing-duration estimates, and disposal-probability models; and advanced optimisation methods such as genetic algorithms, NSGA-II, simulated annealing, or reinforcement learning.

Thus, Chapter 4 moves beyond the present interpretable scheduling baselines and discusses how the framework can be extended toward larger-scale, adaptive, and more realistic judicial scheduling systems.

Chapter 4

Future Work

4.1 From Deterministic Baselines to Richer Scheduling

The experiments in Chapter 3 used deterministic and interpretable scheduling policies. This was an intentional choice. Before attempting advanced optimisation or learning-based techniques, it was necessary to define the scheduling features, construct baseline policies, and understand the trade-offs between age, inactivity gap, case-type fairness, and complexity.

The results showed that these deterministic policies are useful for interpretation. Oldest-first and gap-first policies identify delayed matters, round-robin improves case-type representation, and multi-objective scalarisation exposes the trade-off between delay, fairness, and complexity. However, these methods are still simplified. They operate mainly at the case-selection level and do not fully model judge availability, advocate conflicts, bench compatibility, hearing durations, adjournments, or uncertain disposal.

A realistic court scheduling problem is much harder than the simplified selection problem studied in this thesis. In the current experiments, once a score is assigned to each case, selecting the top cases under a capacity constraint is straightforward. In contrast, a full court scheduling system would involve multiple resources, multiple days, stochastic outcomes, and several hard and soft constraints. Such problems are closely related to resource-constrained scheduling, job-shop scheduling, and multi-objective scheduling, many variants of which are known to be computationally hard or NP-complete (Garey et al. 1976, Błażewicz et al. 1983). Therefore, future work should move from deterministic baselines toward richer heuristic, metaheuristic, and learning-based approaches.

4.2 Richer Stochastic Simulation

The most immediate extension is to build a more realistic stochastic simulator. In the present work, a selected case is removed from the immediate backlog as a first approximation. In reality, a listed case may be heard, adjourned, not reached, withdrawn, dismissed, disposed, or returned to the queue for another hearing.

A future simulator should therefore model the day-of-hearing outcome explicitly. Each listed case may have transition probabilities such as:

$$\Pr(\text{disposed} \mid c), \quad \Pr(\text{adjourned} \mid c), \quad \Pr(\text{not reached} \mid c),$$

where these probabilities may depend on case type, hearing history, order history, bench load, and previous inactivity. Such a simulator would allow the scheduler to be evaluated not only by selected-case composition, but also by long-term backlog evolution.

This extension would also make it possible to study over-listing. Courts may list more matters than can realistically be heard, expecting that some will not proceed. A stochastic simulator can test whether controlled over-listing improves utilisation or instead increases ineffective listings.

4.3 Adding Real-World Constraints

The next extension is to include operational constraints that were recognised but not fully implemented in this thesis. These include judge availability, advocate conflicts, bench-type compatibility, roster constraints, and hearing duration.

For example, if case c requires bench type b_c, then it should only be assigned to compatible benches. Similarly, if the same advocate appears in multiple matters on the same day, the schedule should avoid impossible conflicts. These constraints can be represented as:

$$x_{c,b,t} = 0 \quad \text{if case } c \text{ is not compatible with bench } b,$$

and

$$\sum_{c:a \in A_c} x_{c,t} \leq L_a,$$

where L_a is the feasible daily appearance load for advocate a.

Adding such constraints would make the formulation closer to a true resource-constrained scheduling problem. It would also make the schedules more realistic, because a selected set of cases is only useful if it is feasible with respect to judges, advocates, bench types, and time.

4.4 Genetic Algorithms and Multi-Objective Search

One natural direction for future work is to use genetic algorithms. The scheduling problem can be encoded as a chromosome, where each chromosome represents a candidate schedule or ordered list of cases. A population of such schedules can then be evolved using selection, crossover, and mutation.

The fitness function can combine the objectives studied in this thesis:

$$\text{Fitness}(S) = w_D D(S) + w_F F(S) - w_C C(S) - \text{Penalty}(S),$$

where $D(S)$ measures delay priority, $F(S)$ measures case-type fairness, $C(S)$ measures complexity, and the penalty term captures constraint violations such as over-capacity, bench incompatibility, or advocate conflicts.

A genetic algorithm is suitable because it does not require the objective function to be smooth or convex. It can search large combinatorial spaces and handle multiple constraints through penalty terms. However, a simple genetic algorithm still requires the weights in the fitness function to be chosen manually.

A stronger extension would be to use a multi-objective genetic algorithm such as NSGA-II (Deb et al. 2002). Instead of producing one schedule, NSGA-II can approximate a Pareto frontier of schedules. One schedule may prioritise delay reduction, another may prioritise fairness, and another may reduce complexity. This is attractive for judicial scheduling because there may not be a single universally best schedule. Different courts or administrative settings may prefer different points on the Pareto frontier.

4.5 Reinforcement Learning Framework

Reinforcement learning is another possible future direction, but it should be used only after a credible simulator is available. In reinforcement learning, the scheduler can be modelled as an agent that interacts with a court environment over time (Sutton & Barto 2018).

The state at day t may include the current backlog, case ages, inactivity gaps, hearing histories, case-type distribution, bench availability, and advocate availability. The action is the selection of cases to list on that day. The environment then simulates what happens: some matters are heard, some are adjourned, some are not reached, and some are disposed.

A possible reward function is:

$$R_t = \lambda_1(\text{disposals}) - \lambda_2(\text{old backlog}) - \lambda_3(\text{fairness error}) - \lambda_4(\text{ineffective listings}).$$

The aim of the agent is to learn a policy that maximises long-term reward, not only immediate selected-case quality. This is useful because court scheduling is inherently sequential. A decision made today changes the backlog available tomorrow.

However, reinforcement learning also has risks. If the simulator is unrealistic, the learned policy may exploit artificial patterns and fail in practice. Therefore, RL should be treated as a later-stage extension, after deterministic baselines, stochastic simulation, and constraint-aware optimisation have been developed.

4.6 Prediction-Assisted Scheduling

Another practical extension is to combine supervised prediction with scheduling. Instead of directly learning a scheduling policy, machine learning models can first estimate useful quantities such as hearing duration, adjournment probability, disposal probability, or probability of being not reached.

For example, a model may estimate:

$$\widehat{q}_c = \Pr(\text{disposed if listed} \mid c),$$

or

$$\widehat{d_c} = \text{expected hearing duration of case } c.$$

These predictions can then be inserted into an optimisation or heuristic scheduling model. This hybrid approach may be more practical than end-to-end reinforcement learning because the prediction task is easier to validate. It also keeps the final scheduler interpretable.

4.7 Practical and Institutional Considerations

Any real deployment of a scheduling system in the judiciary would require careful institutional design. The model should not replace judicial discretion. Instead, it should act as a decision-support tool that highlights delayed matters, detects under-represented categories, and suggests possible cause-list compositions.

Transparency is essential. Every recommendation should be explainable in terms of observable features such as age, inactivity gap, hearing history, or fairness requirement. The system should also allow human override, because legal urgency and procedural context may not always be captured in the data.

Data quality is another major concern. Missing orders, inconsistent case-status information, incomplete hearing histories, or errors in scraped data can affect the schedule. Therefore, future work should include validation pipelines, data-quality checks, and mechanisms for handling missing or uncertain information.

4.8 Summary

This thesis begins with interpretable deterministic baselines because they make the scheduling trade-offs visible. The natural future direction is to make the model more realistic and more powerful. A richer simulator can model uncertain hearing outcomes. Constraint-aware optimisation can incorporate judges, advocates, benches, and hearing durations. Genetic algorithms and NSGA-II can search large multi-objective schedule spaces. Reinforcement learning can be explored once a reliable sequential simulator exists. Prediction-assisted scheduling can estimate disposal and duration signals while preserving interpretability.

Thus, the future of this work lies in moving from a transparent case-selection model toward a realistic, adaptive, and constraint-aware judicial scheduling framework.

Chapter 5

Conclusion

5.1 Conclusion

This thesis studied case listing in the Indian judiciary as a scheduling problem. The work began from the observation that judicial delay is not only a question of total case volume, but also a question of how limited listing opportunities are allocated across pending matters. Using arbitration-related cases from the Appellate Side of the Calcutta High Court, the thesis developed a feature-based abstraction in which cases are represented through age, inactivity gap, hearing-history count, order count, and case type.

The main contribution of the thesis is the conversion of a descriptive court-delay problem into a mathematical and algorithmic scheduling framework. The work first formulated the case-selection problem under limited capacity, then compared deterministic and interpretable scheduling policies. The experiments showed that oldest-first, gap-first, weighted age-gap, balanced-priority, SJF-proxy, and round-robin policies each reveal a different aspect of the listing problem.

The results also showed that no single objective is sufficient. A pure delay-priority rule can select old and inactive cases effectively, but may ignore smaller case categories such as ADCOM. A fairness-oriented rule improves representation, but may reduce average delay priority. A complexity-oriented rule can select simpler matters, but may leave older cases behind. This confirms the central mathematical insight of the thesis: court scheduling should be treated as a multi-objective decision problem rather than as a single-score ranking task.

The bi-objective and multi-objective experiments made these trade-offs explicit. Delay, fairness, and complexity do not always move in the same direction. These findings suggest that algorithmic scheduling can be useful not because it produces one universally correct schedule, but because it exposes the consequences of different scheduling priorities.

At the same time, the thesis does not claim that the proposed scheduler is ready for deployment. The present implementation is a first-stage abstraction. It operates at the selection level and does not fully model judge availability, advocate conflicts, bench compatibility, hearing duration, adjournment probability, or disposal probability. Real court scheduling is affected by procedural, administrative, and institutional realities that cannot be captured completely through the current dataset.

The broader implication is that improvement in judicial scheduling will require both algorithmic and operational changes. From the algorithmic side, richer models can incorporate stochastic hearing outcomes, multi-day backlog evolution, bench constraints, advocate availability, and predictive estimates of disposal or adjournment. From the operational side, courts would benefit from more structured digital records, standardised hearing-outcome labels, reliable case-status updates, machine-readable order metadata, and better tracking of listing outcomes such as "heard", "adjourned", "not reached", or "disposed".

In practice, a scheduling system for courts should be designed as a decision-support tool rather than as a replacement for judicial discretion. It can highlight long-inactive matters, detect underrepresented case categories, estimate likely scheduling conflicts, and suggest balanced cause-list compositions. Human review must remain central, especially where legal urgency, procedural context, or judicial judgement cannot be reduced to numerical features.

Thus, the thesis demonstrates that there is clear scope for improving court scheduling mathematically and algorithmically. The present work provides a first structured step: it identifies relevant scheduling features, defines interpretable policies, compares single-objective and multi-objective formulations, and shows where trade-offs arise. Future work can build on this foundation to develop more realistic, adaptive, and institutionally usable scheduling systems for the Indian judiciary.

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